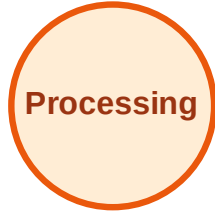


BFS Traversal

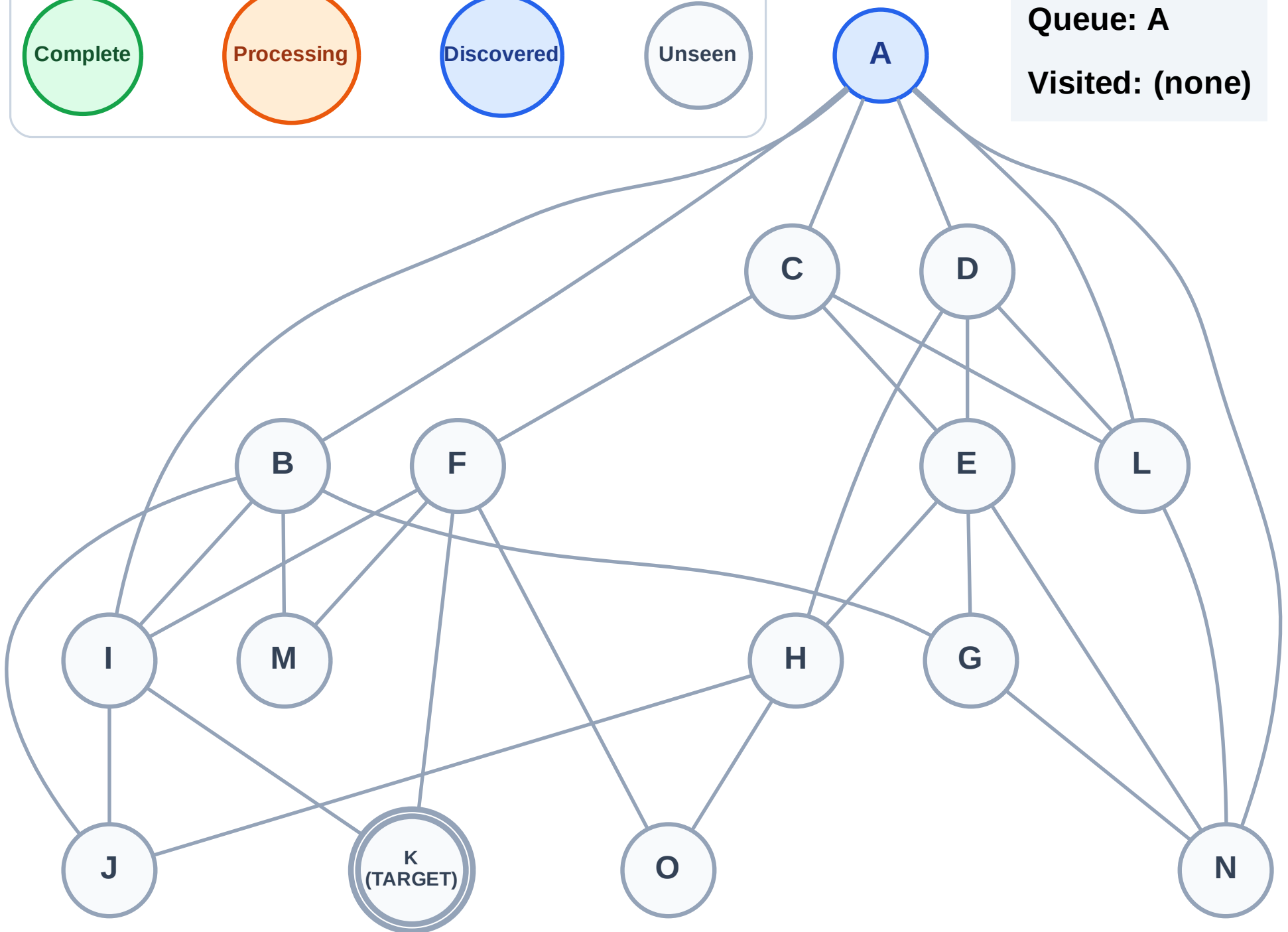
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

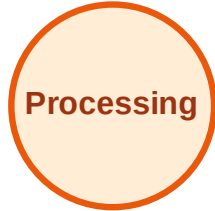
Visited: (none)



BFS Traversal

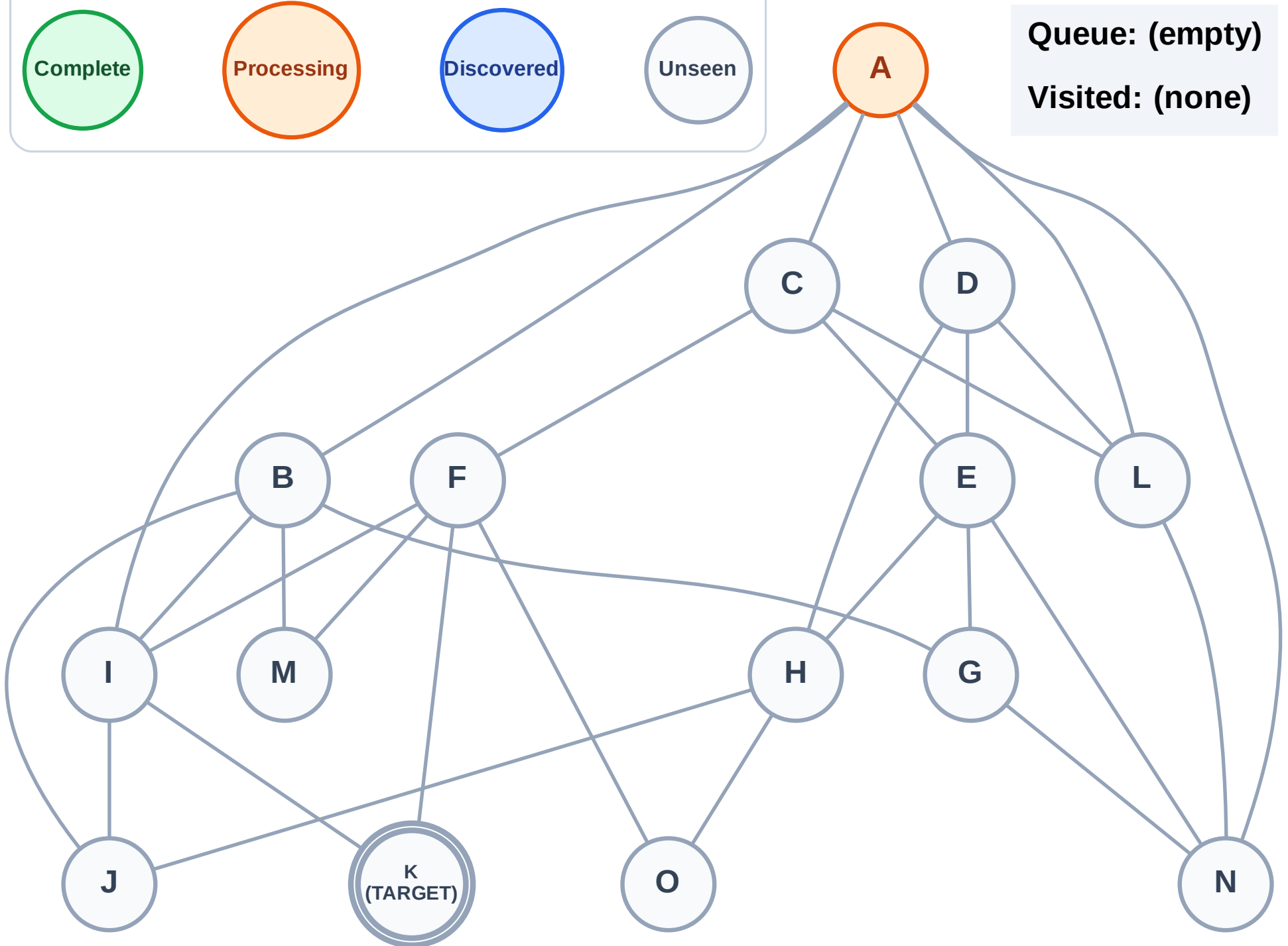
Step 2: Process node A

Legend



Queue: (empty)

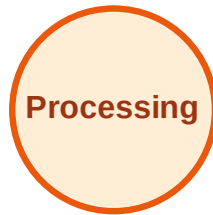
Visited: (none)



BFS Traversal

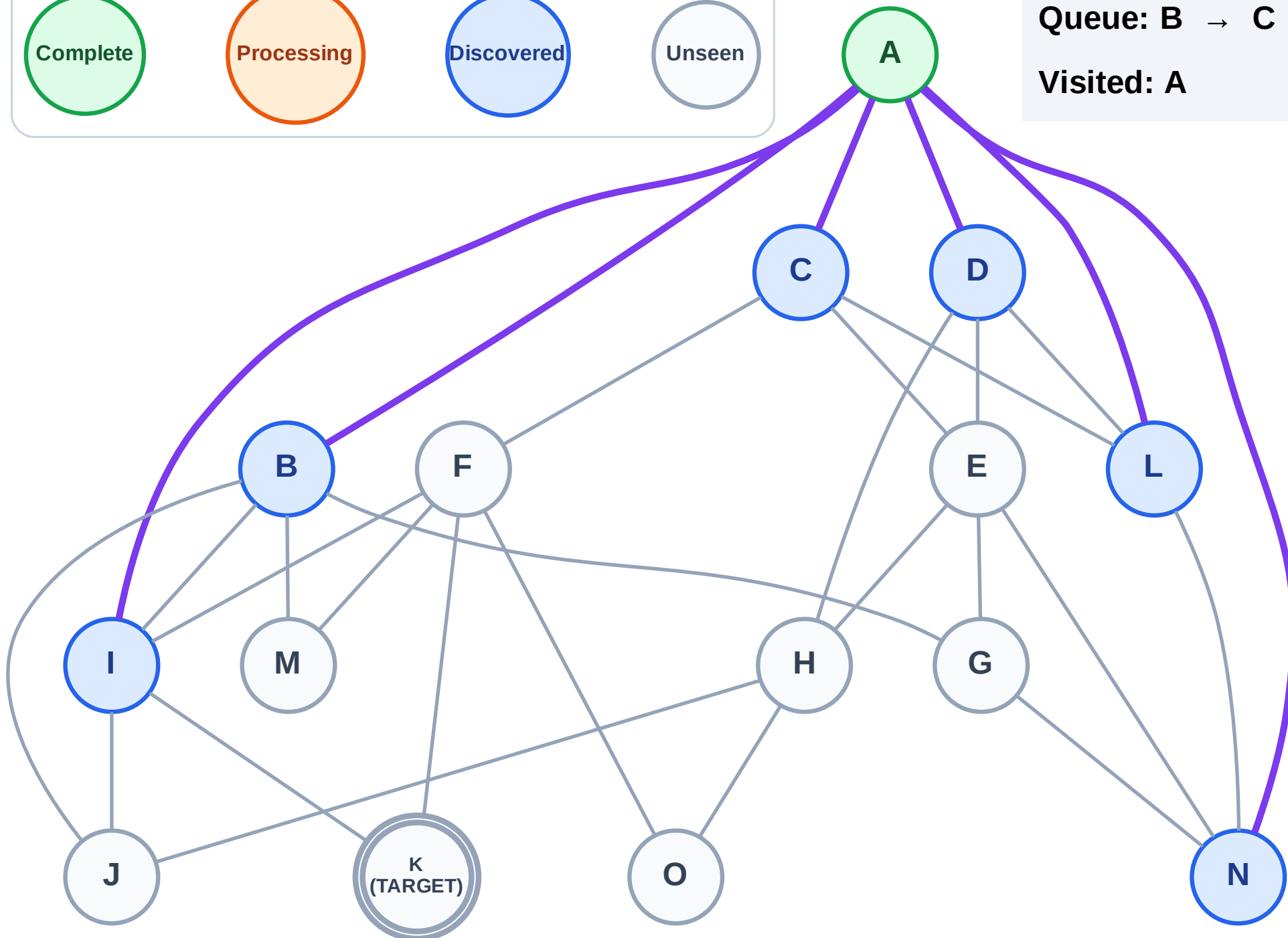
Step 3: Discover B, C, D, I, L, N from A

Legend



Queue: B → C → D → I → L → N

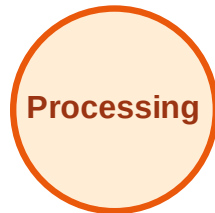
Visited: A



BFS Traversal

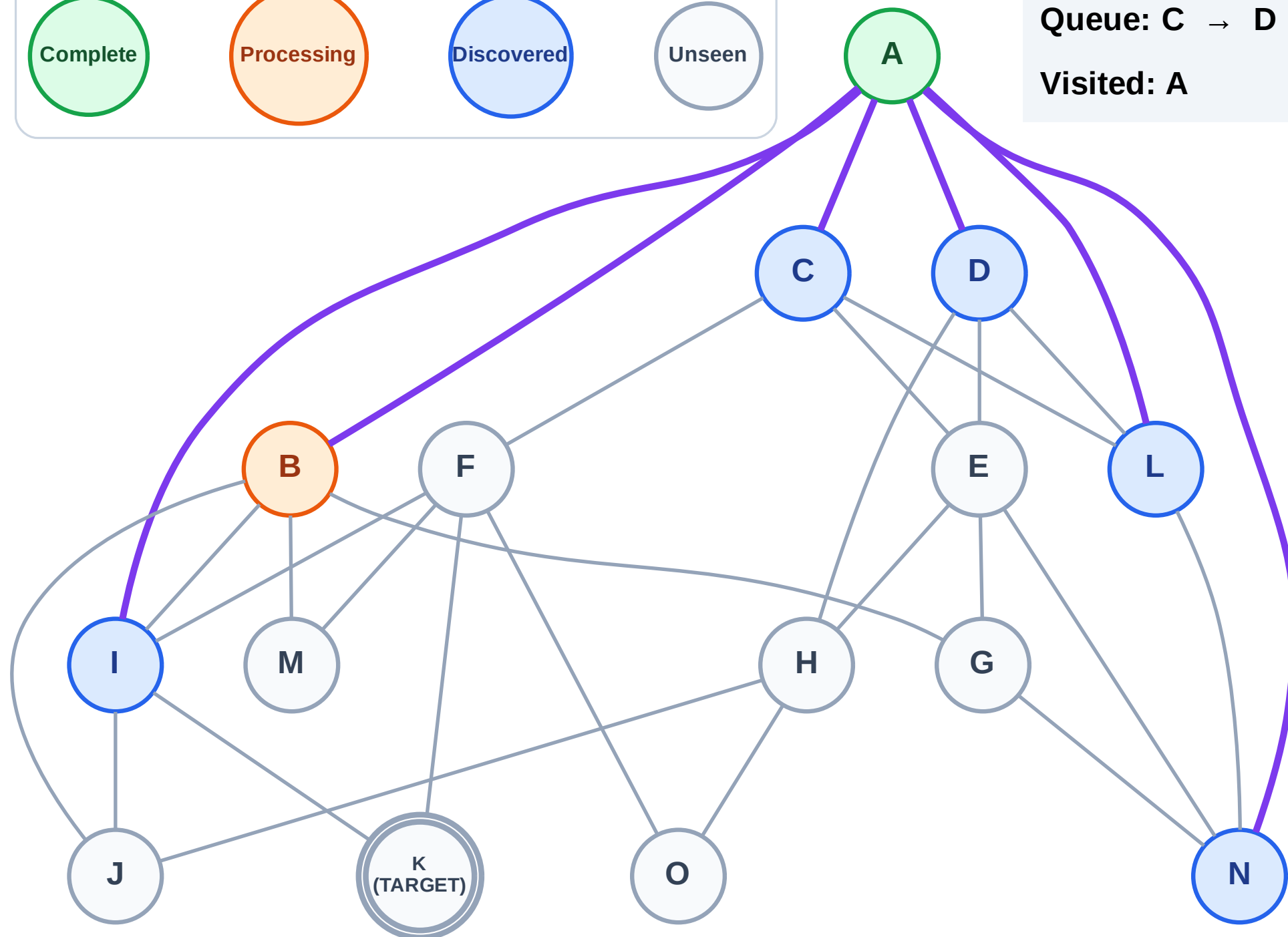
Step 4: Process node B

Legend



Queue: C → D → I → L → N

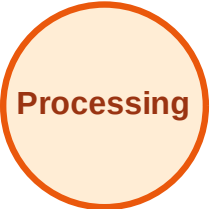
Visited: A



BFS Traversal

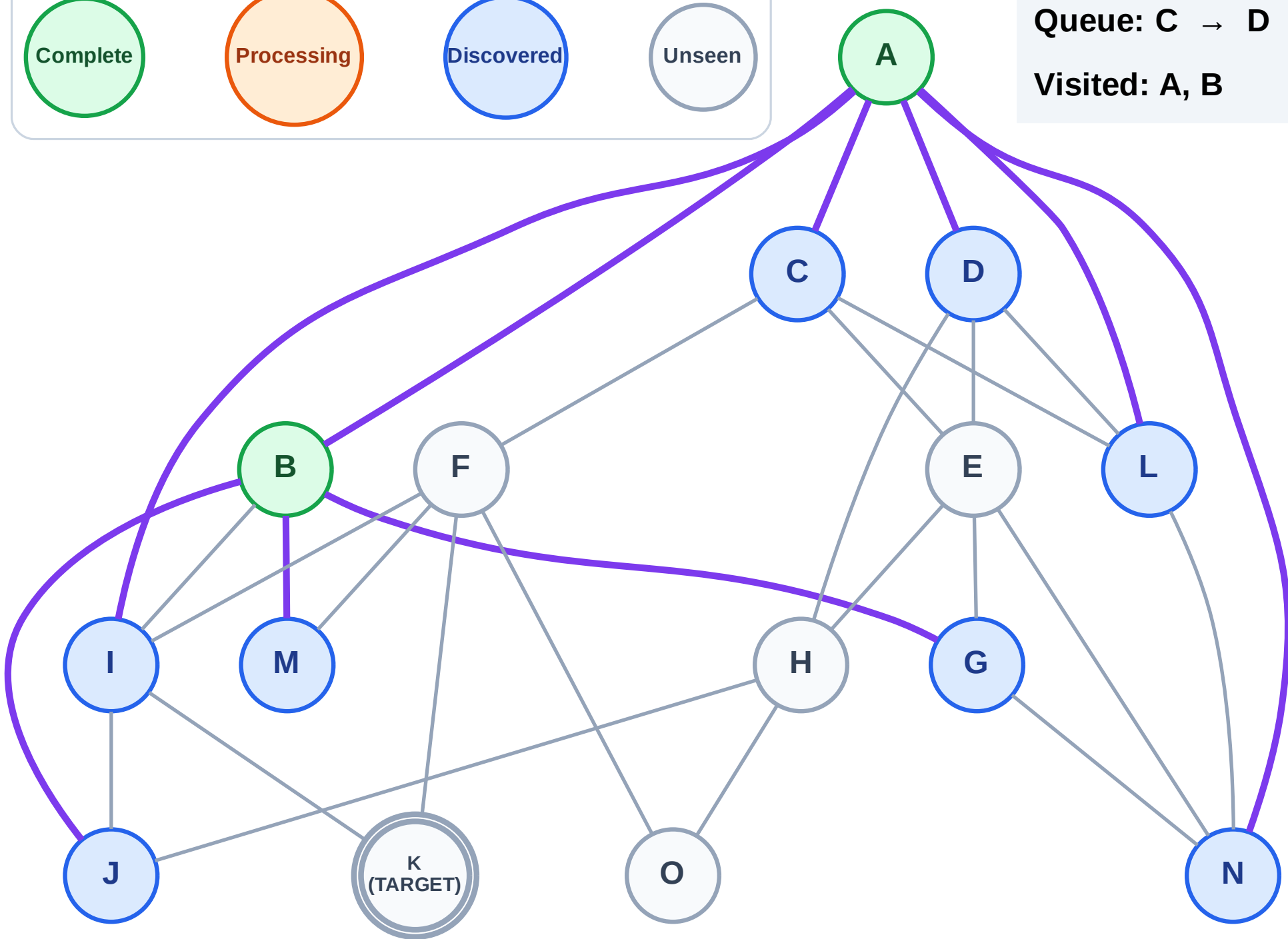
Step 5: Discover G, J, M from B

Legend



Queue: C → D → I → L → N → G → J → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

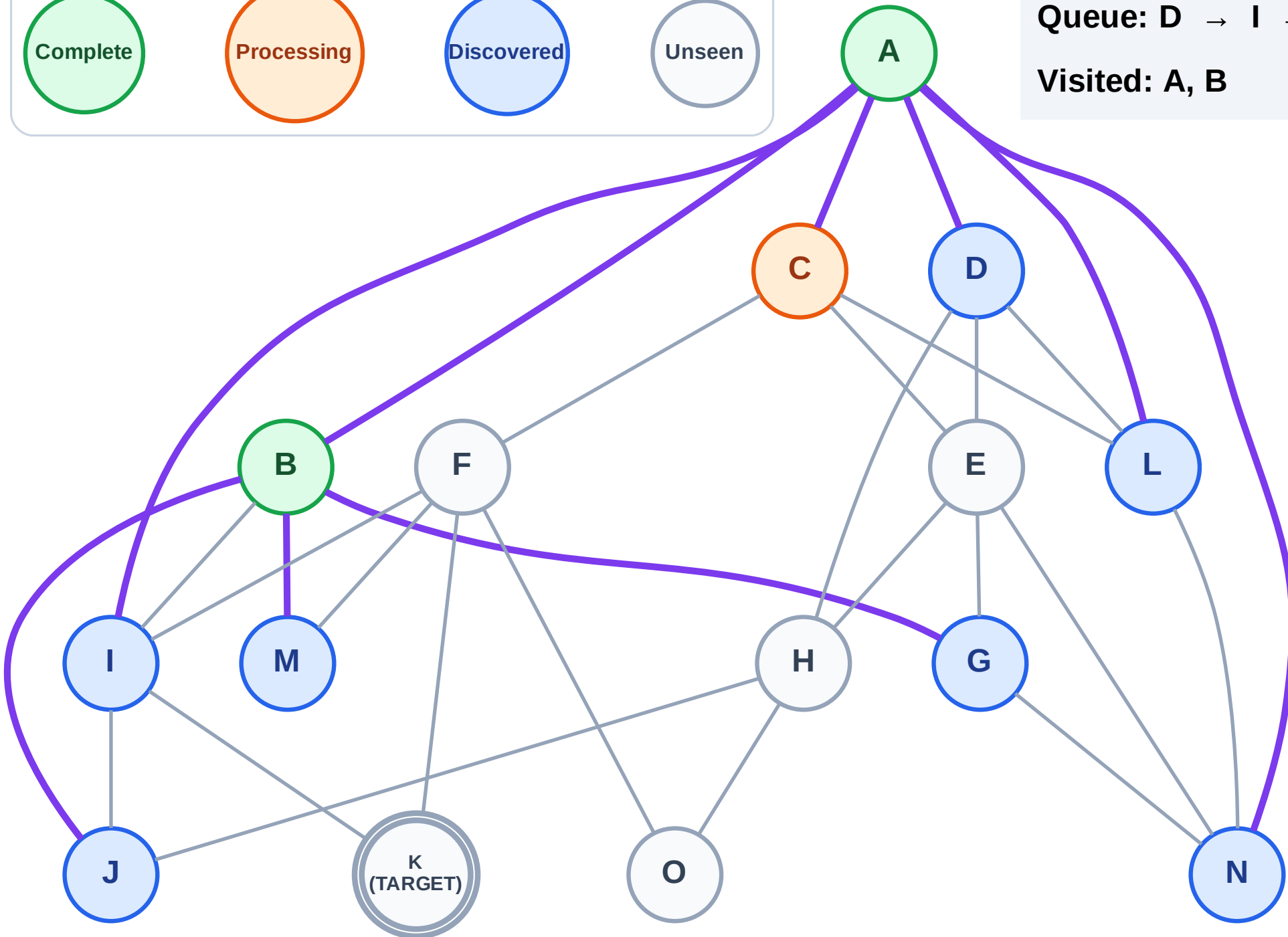
Processing

Discovered

Unseen

Queue: D → I → L → N → G → J → M

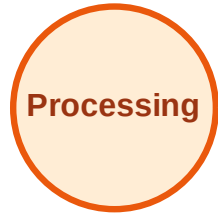
Visited: A, B



BFS Traversal

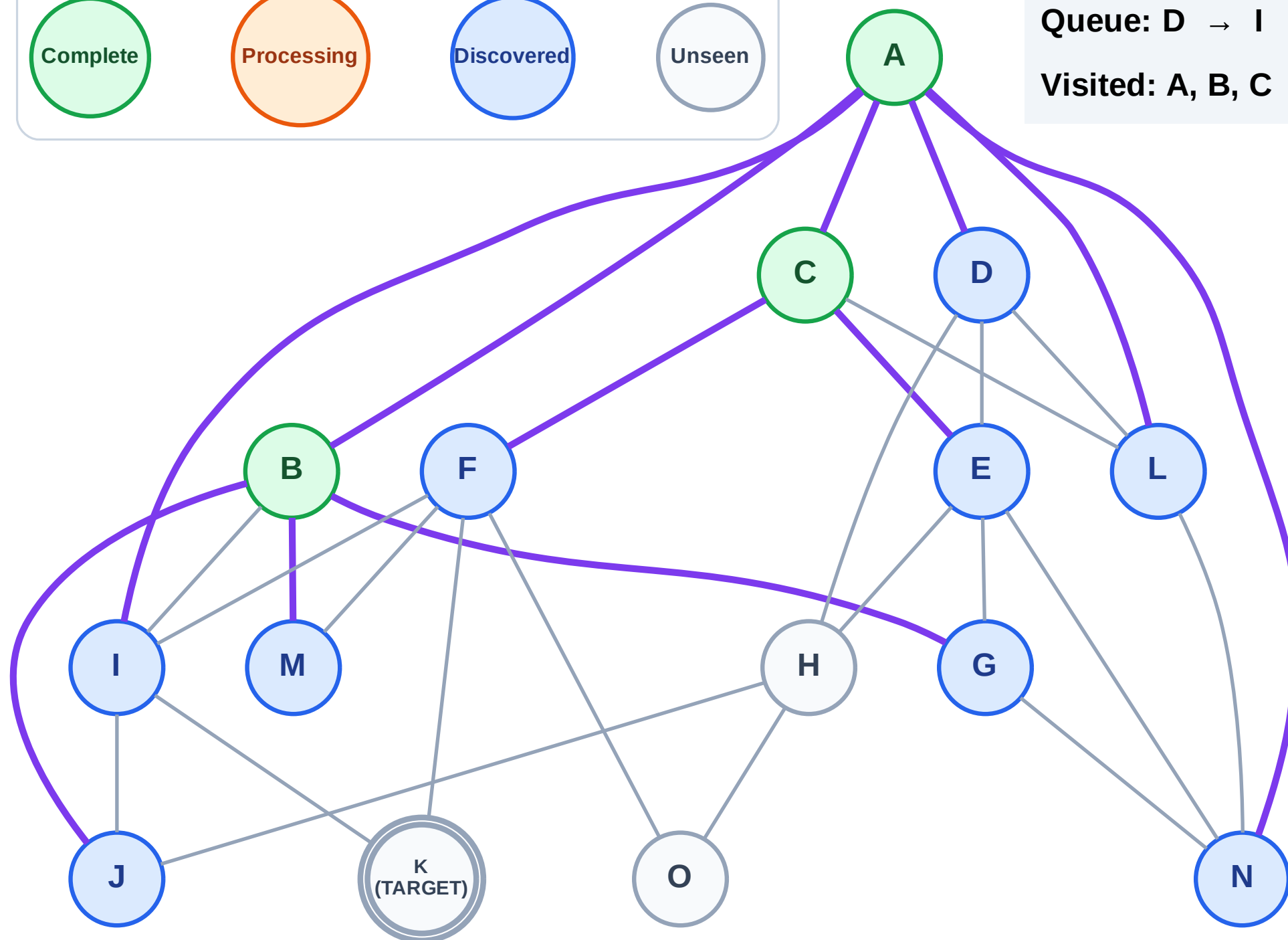
Step 7: Discover E, F from C

Legend



Queue: D → I → L → N → G → J → M → E → F

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

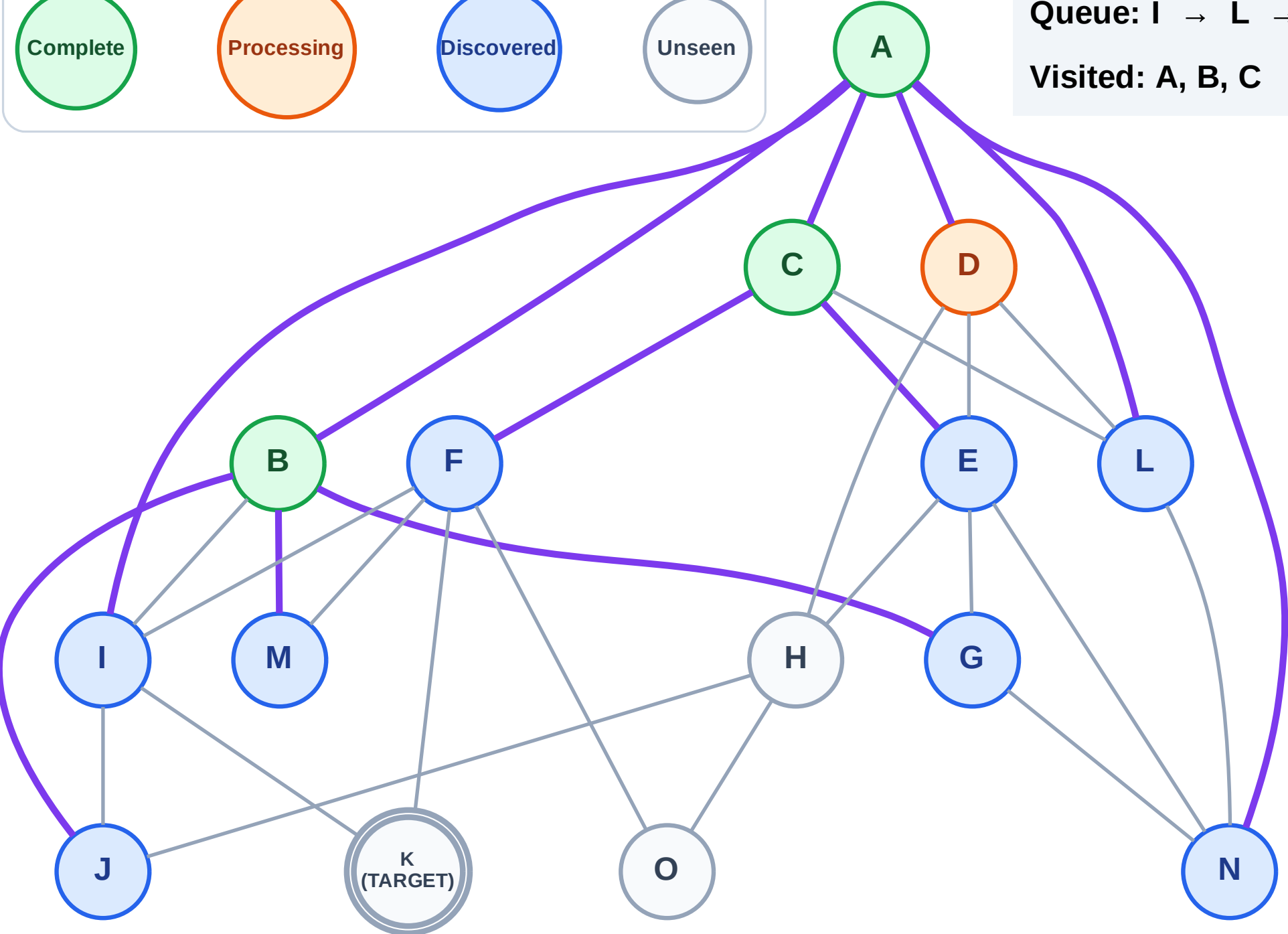
Processing

Discovered

Unseen

Queue: I → L → N → G → J → M → E → F

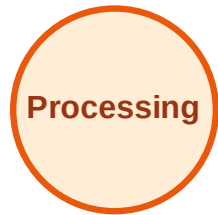
Visited: A, B, C



BFS Traversal

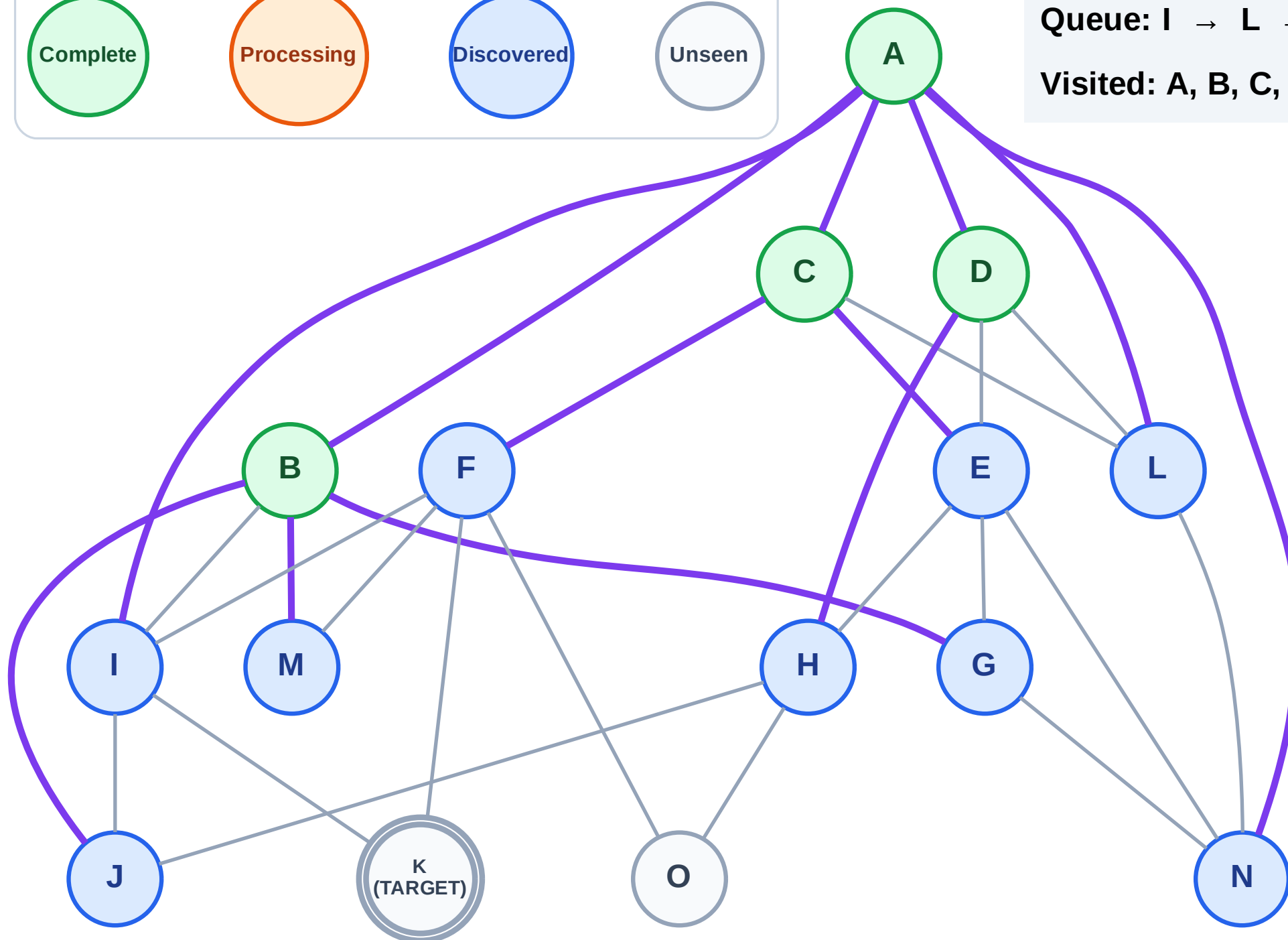
Step 9: Discover H from D

Legend



Queue: I → L → N → G → J → M → E → F → H

Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

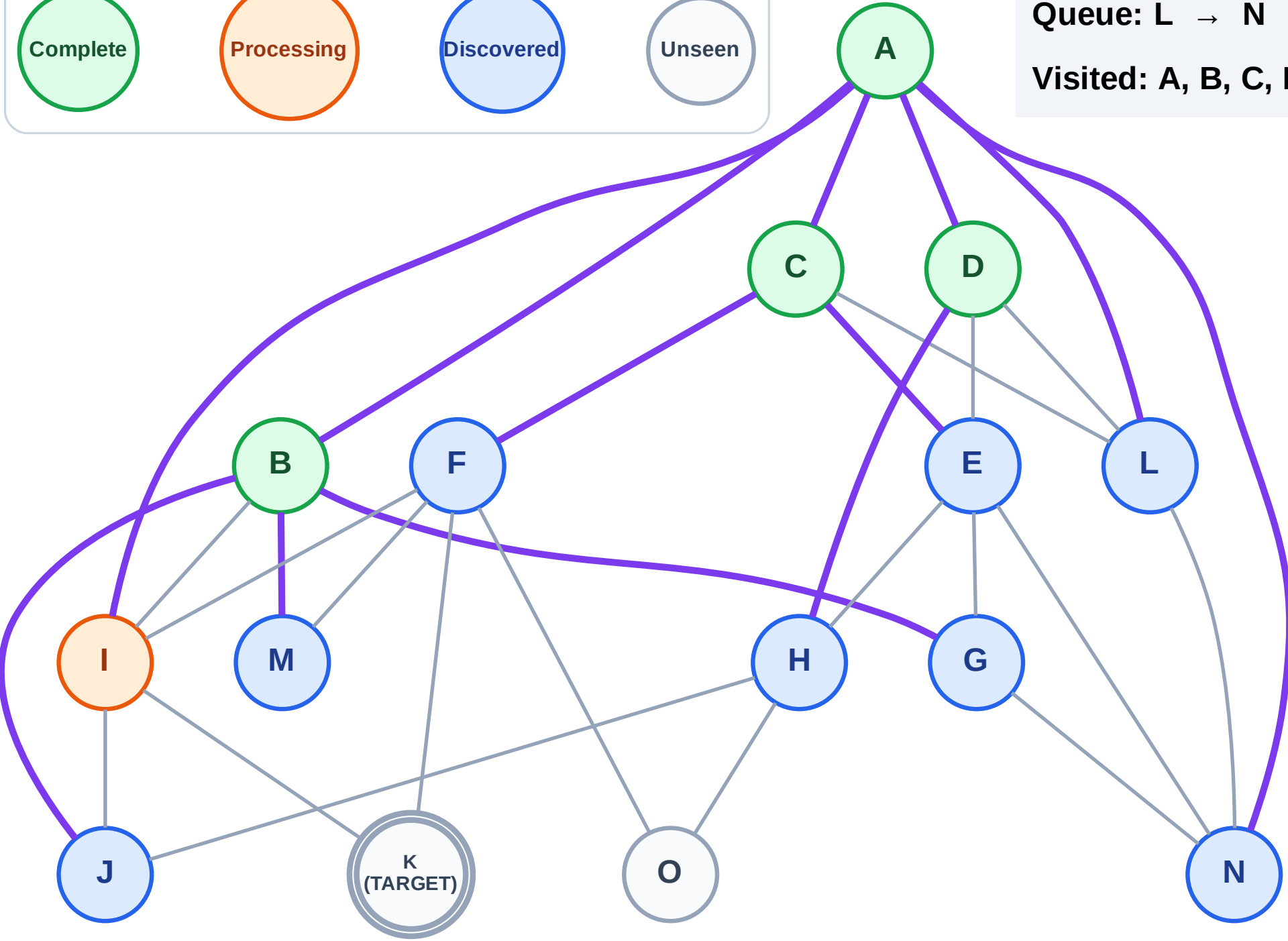
Processing

Discovered

Unseen

Queue: L → N → G → J → M → E → F → H

Visited: A, B, C, D



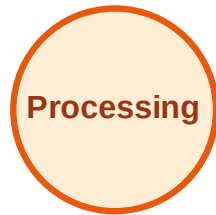
BFS Traversal

Step 11: Discover K from I

Legend



Complete



Processing



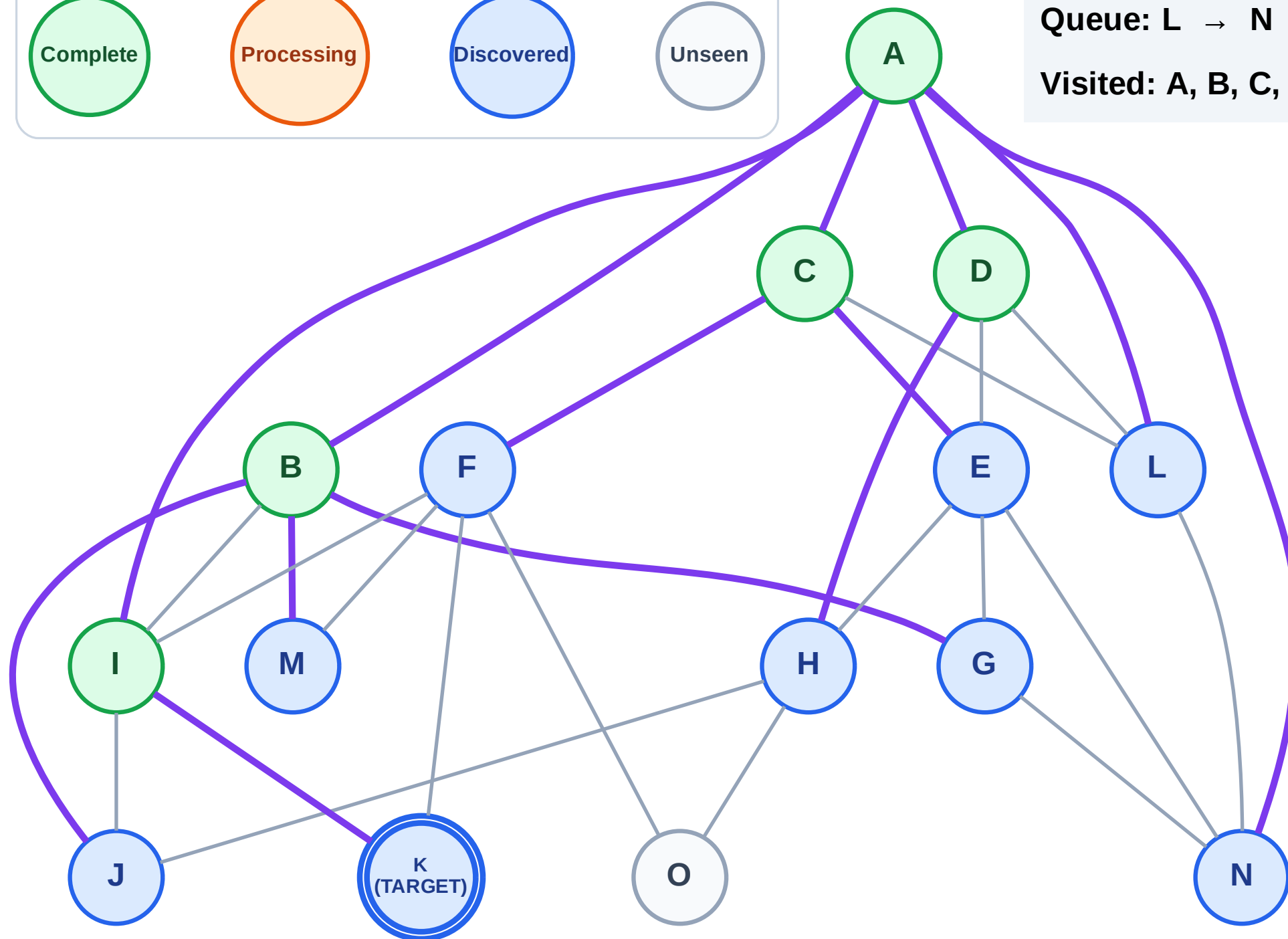
Discovered



Unseen

Queue: L → N → G → J → M → E → F → H → K

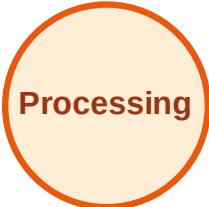
Visited: A, B, C, D, I



BFS Traversal

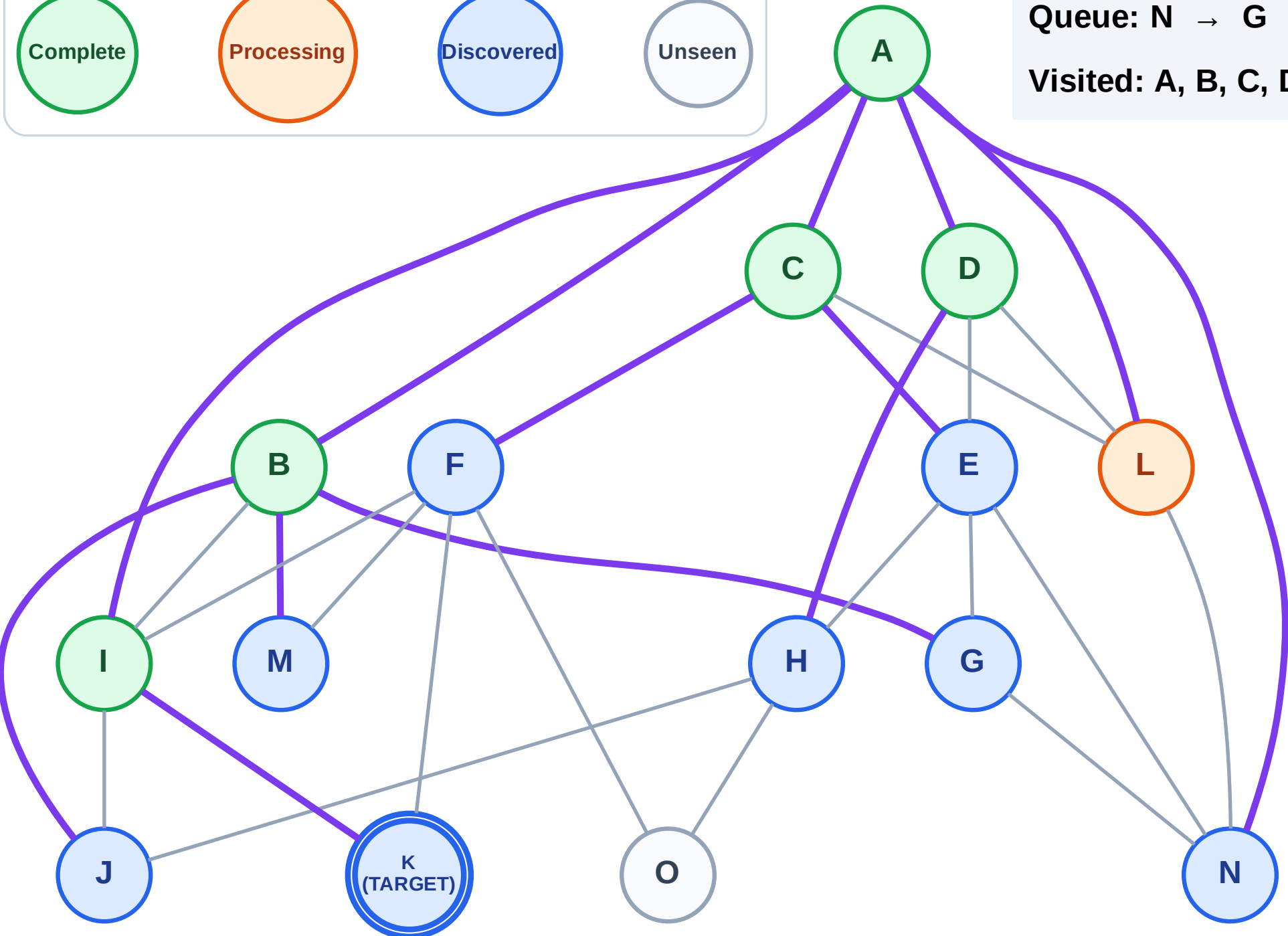
Step 12: Process node L

Legend



Queue: N → G → J → M → E → F → H → K

Visited: A, B, C, D, I



BFS Traversal

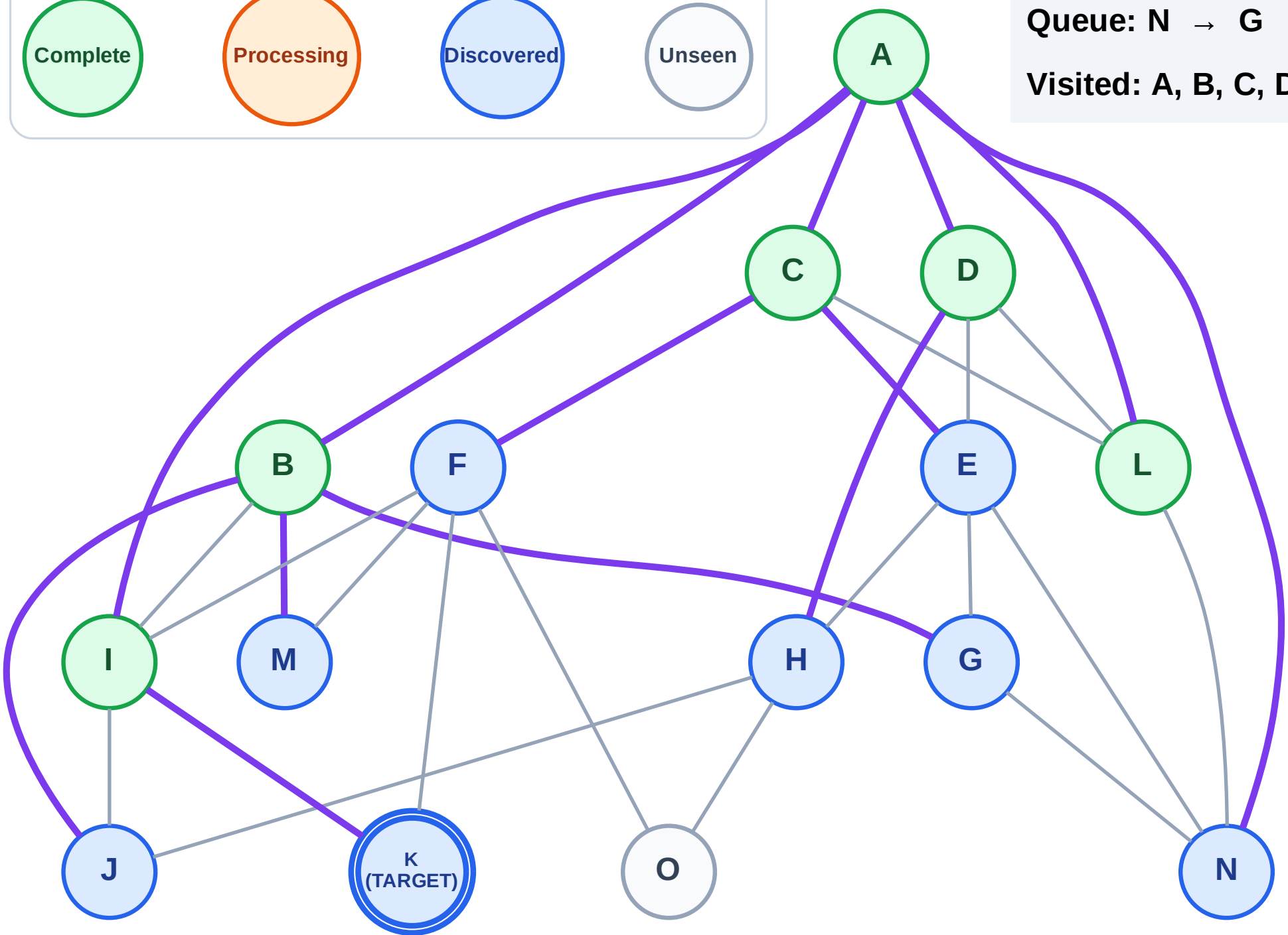
Step 13: No new neighbors from L

Legend



Queue: N → G → J → M → E → F → H → K

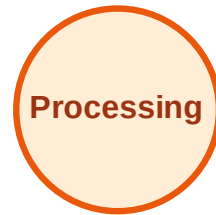
Visited: A, B, C, D, I, L



BFS Traversal

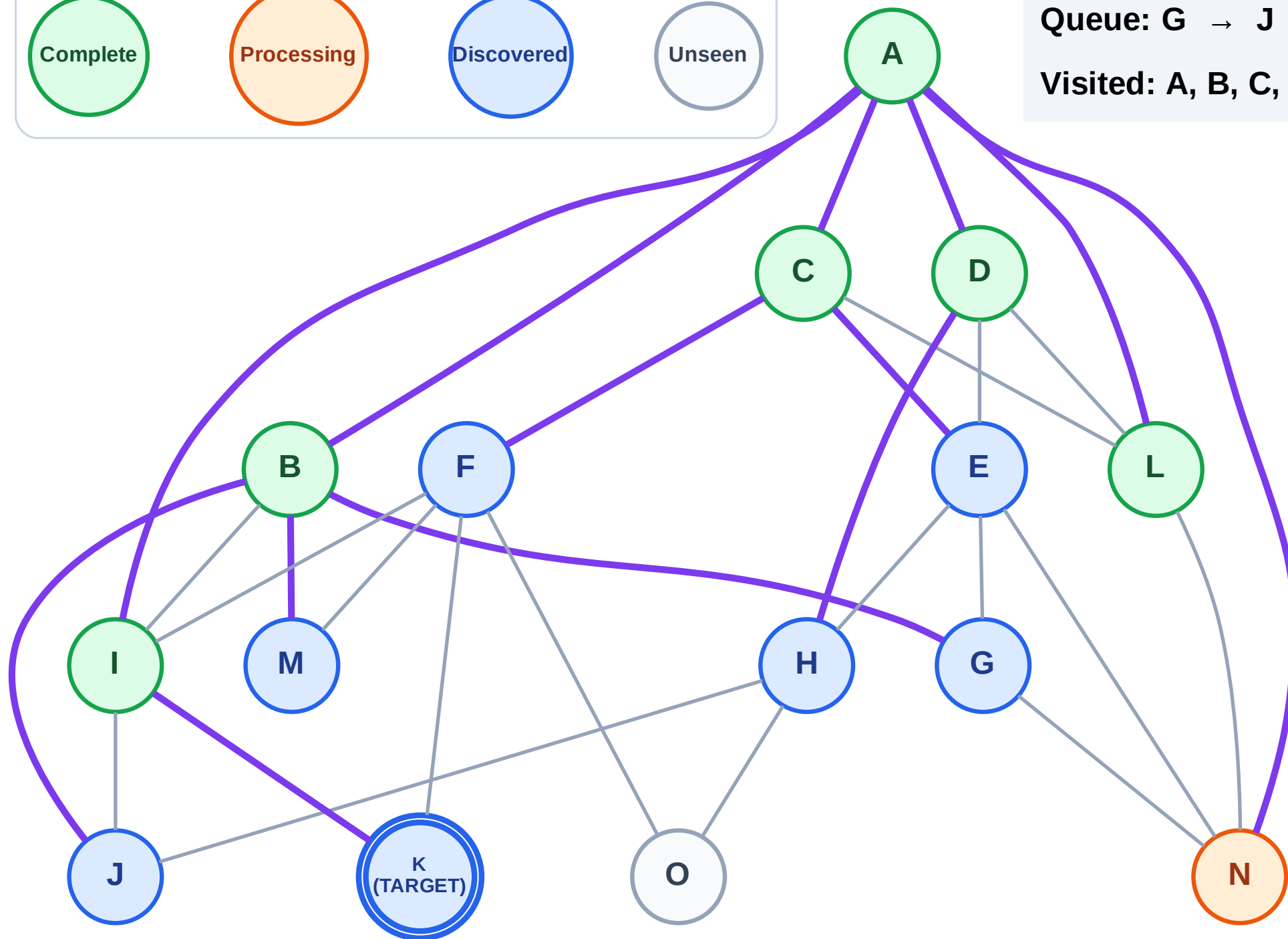
Step 14: Process node N

Legend



Queue: G → J → M → E → F → H → K

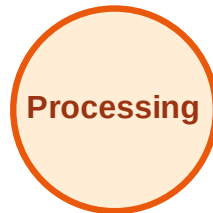
Visited: A, B, C, D, I, L



BFS Traversal

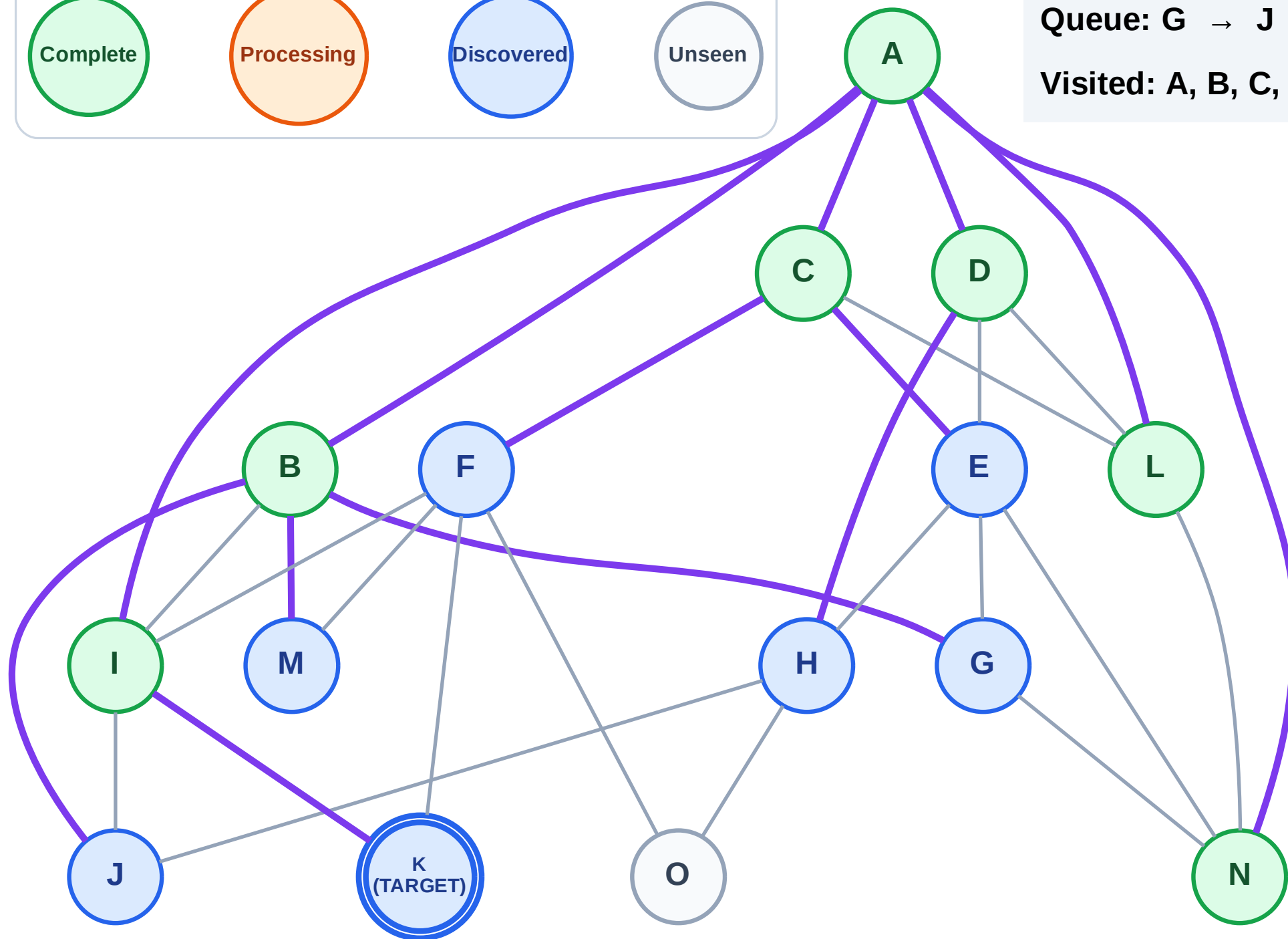
Step 15: No new neighbors from N

Legend



Queue: G → J → M → E → F → H → K

Visited: A, B, C, D, I, L, N



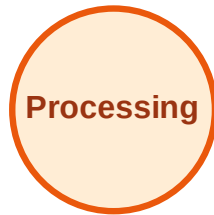
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



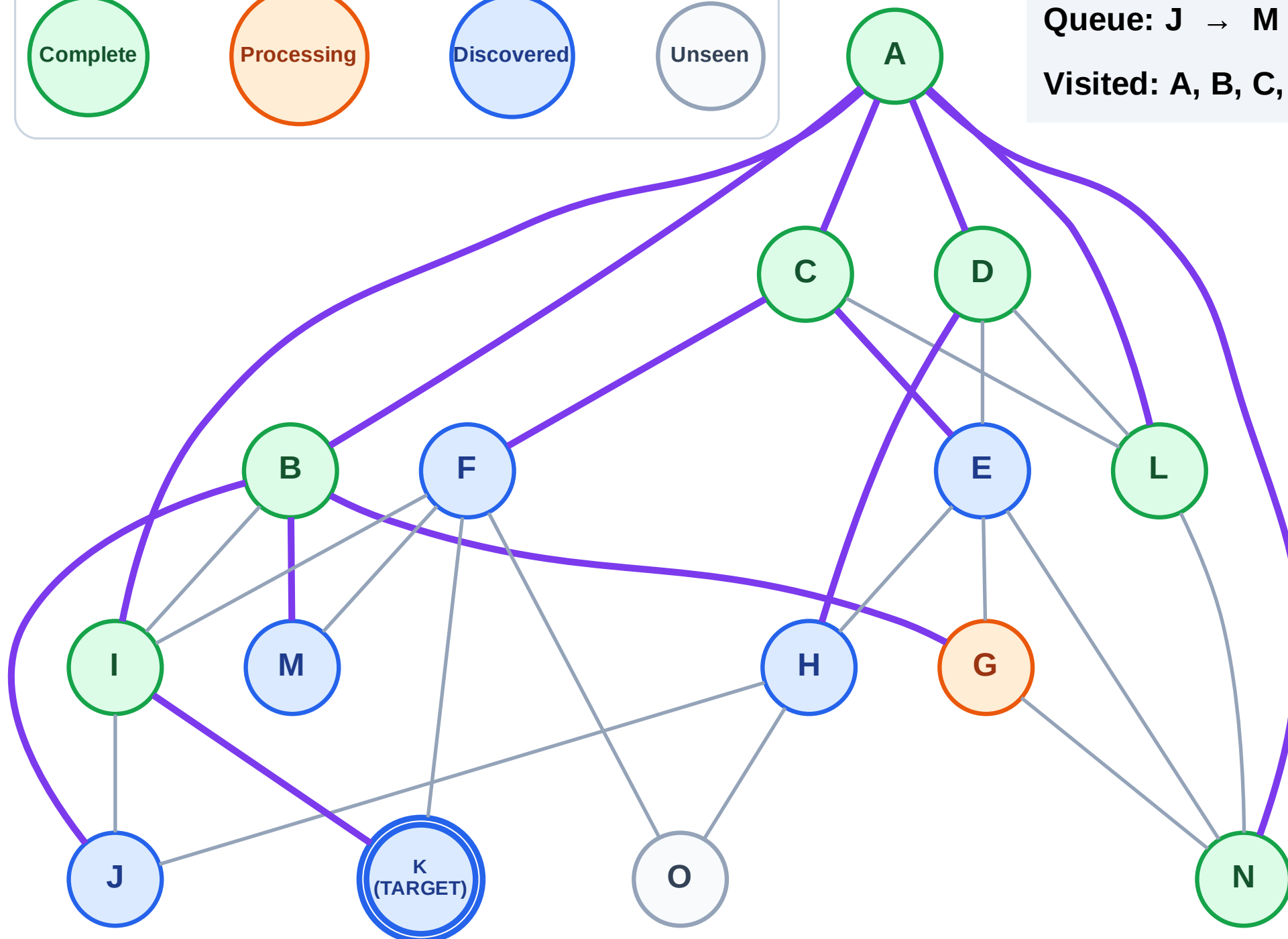
Discovered



Unseen

Queue: J → M → E → F → H → K

Visited: A, B, C, D, I, L, N



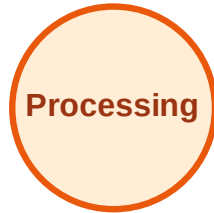
BFS Traversal

Step 17: No new neighbors from G

Legend



Complete



Processing



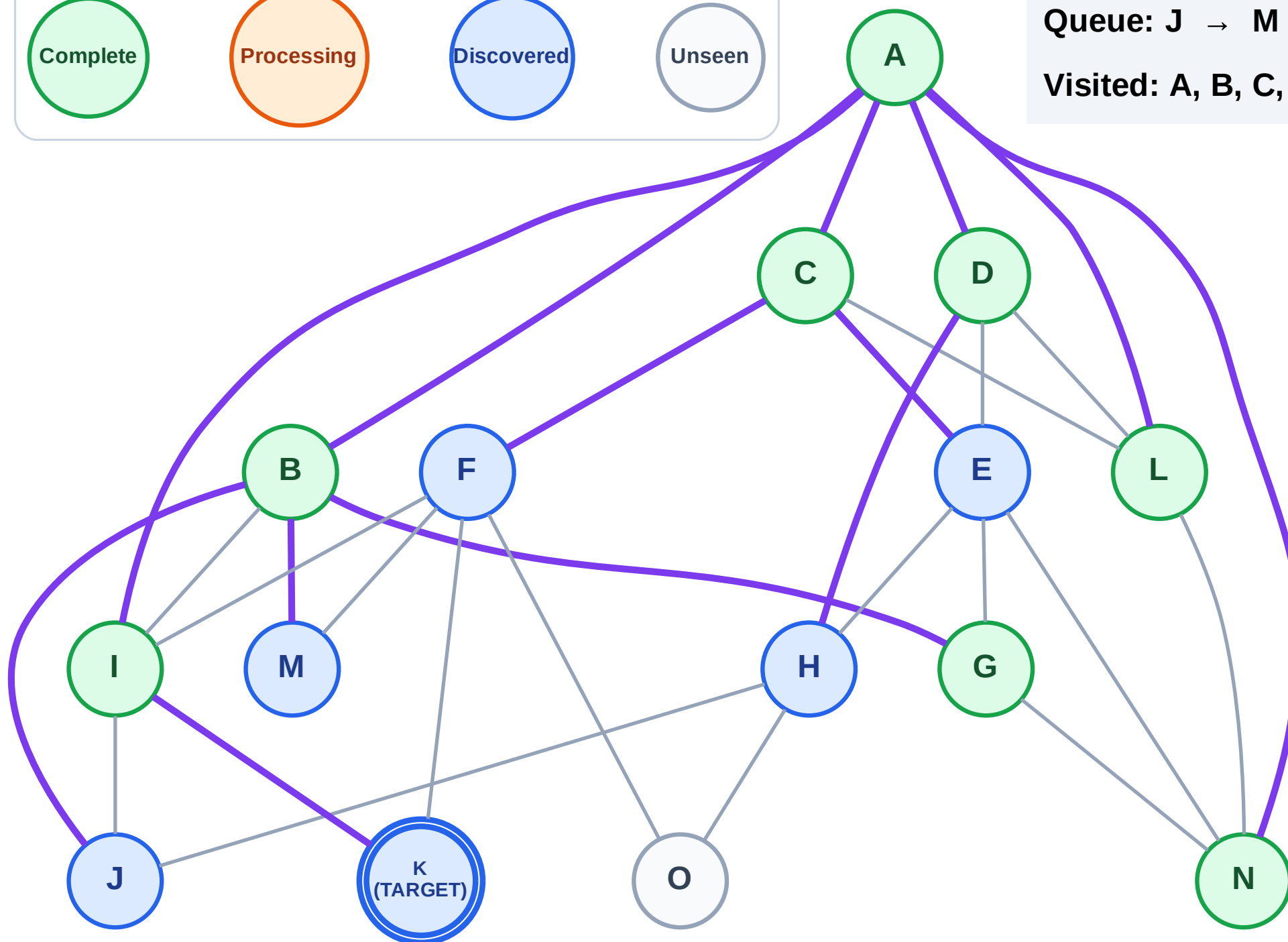
Discovered



Unseen

Queue: J → M → E → F → H → K

Visited: A, B, C, D, G, I, L, N



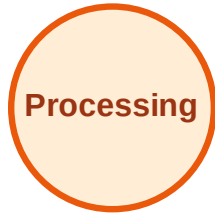
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



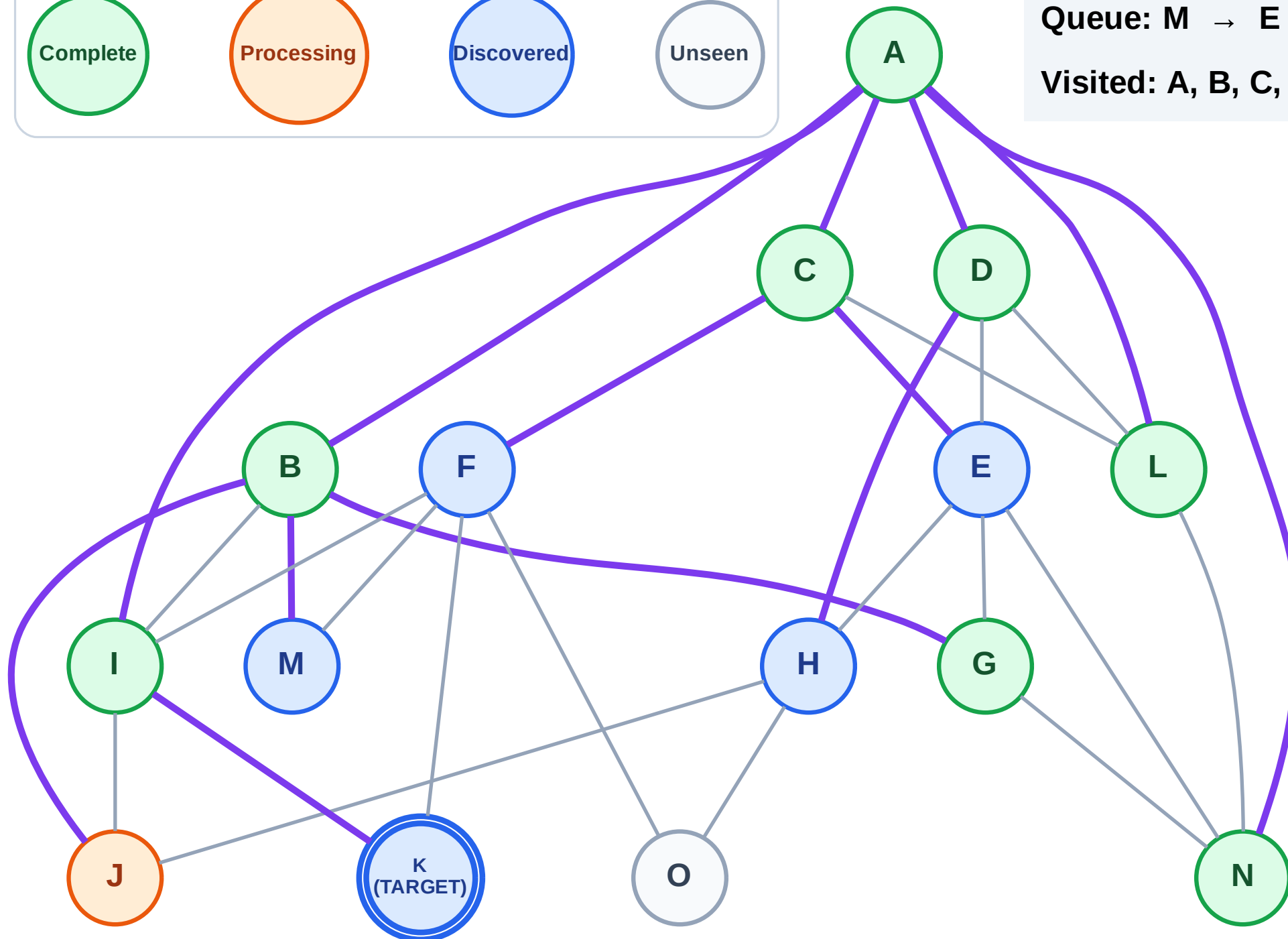
Discovered



Unseen

Queue: M → E → F → H → K

Visited: A, B, C, D, G, I, L, N



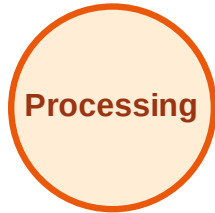
BFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



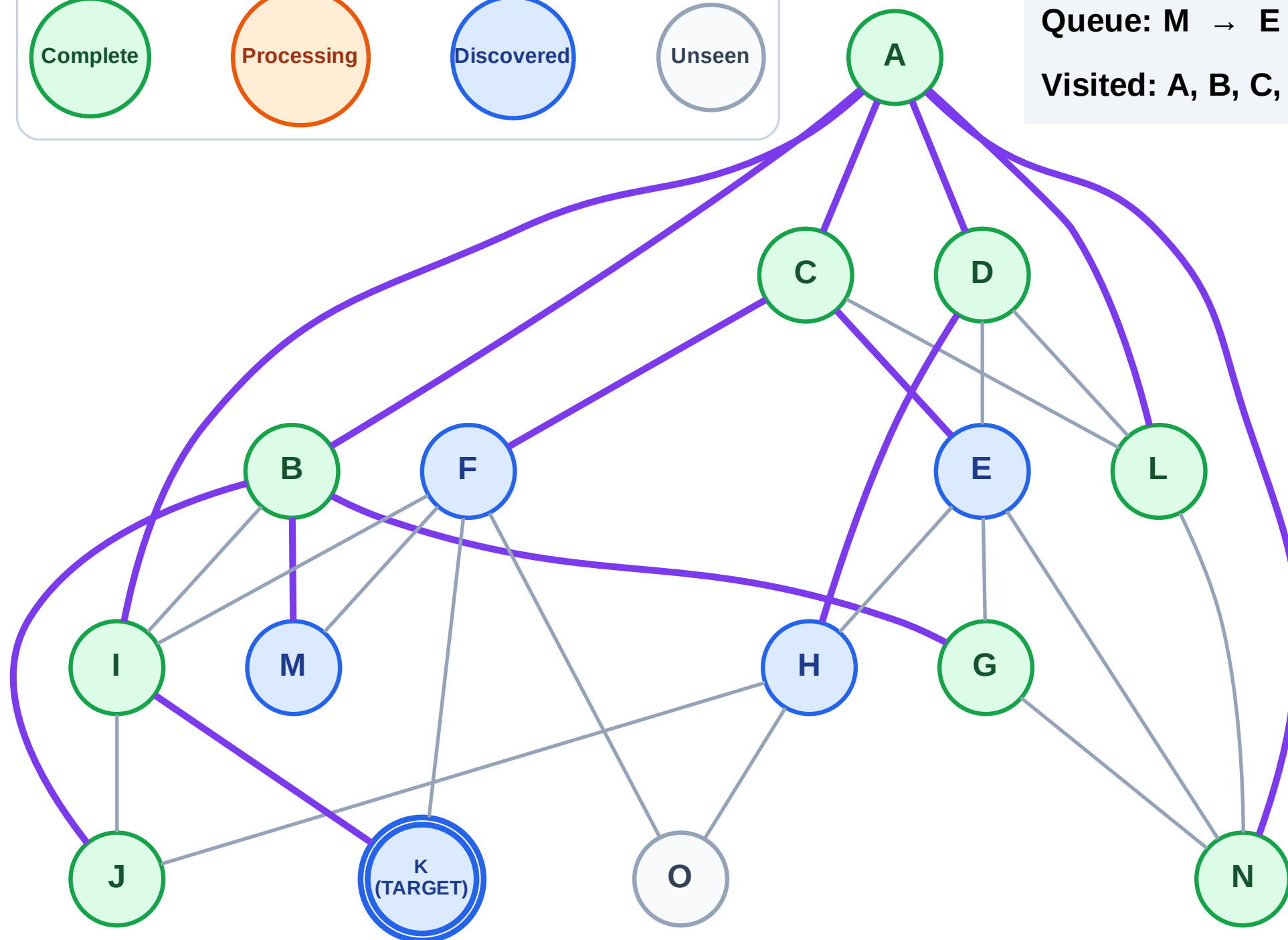
Discovered



Unseen

Queue: M → E → F → H → K

Visited: A, B, C, D, G, I, J, L, N



BFS Traversal

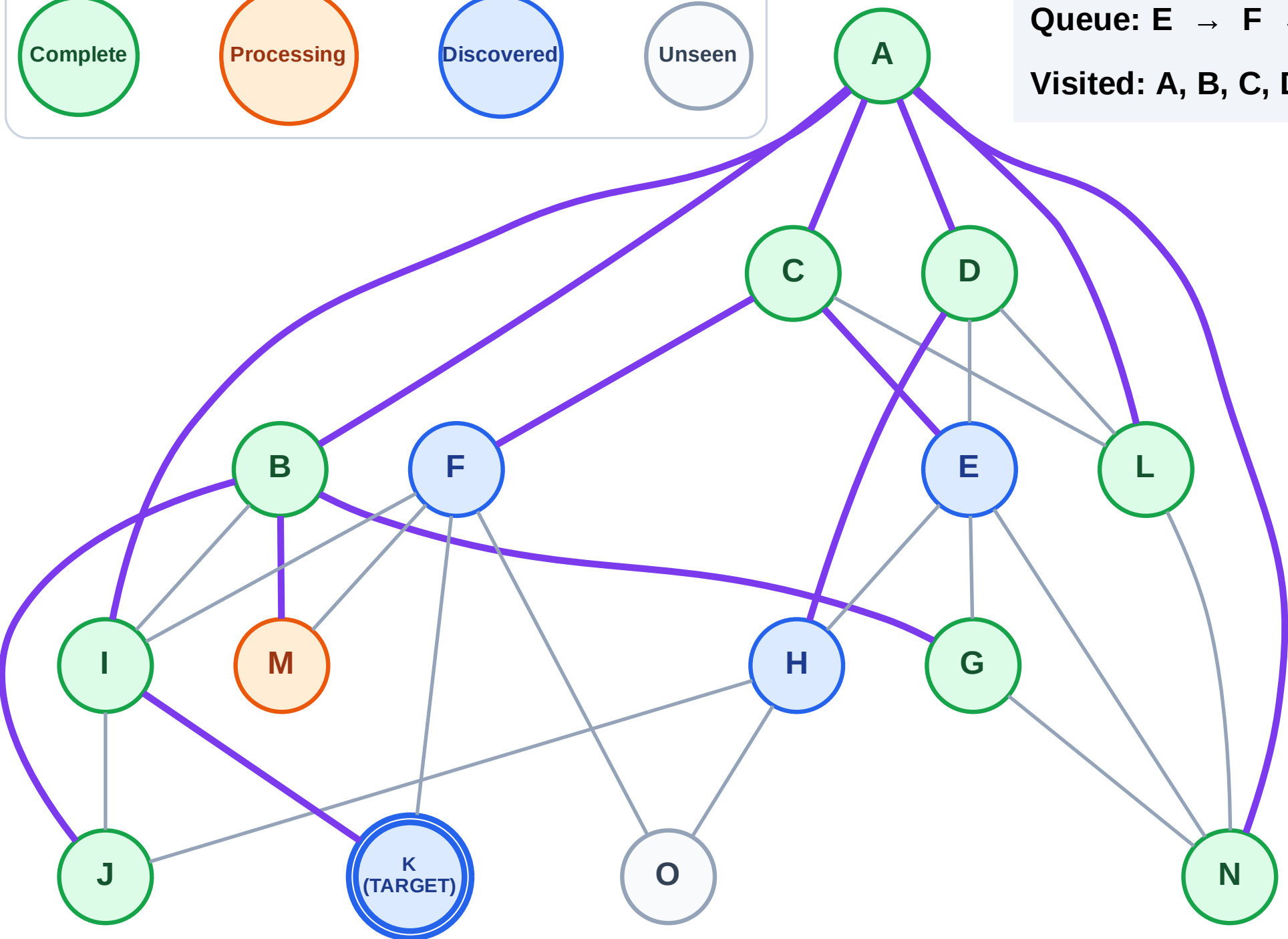
Step 20: Process node M

Legend



Queue: E → F → H → K

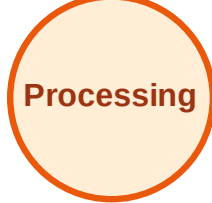
Visited: A, B, C, D, G, I, J, L, N



BFS Traversal

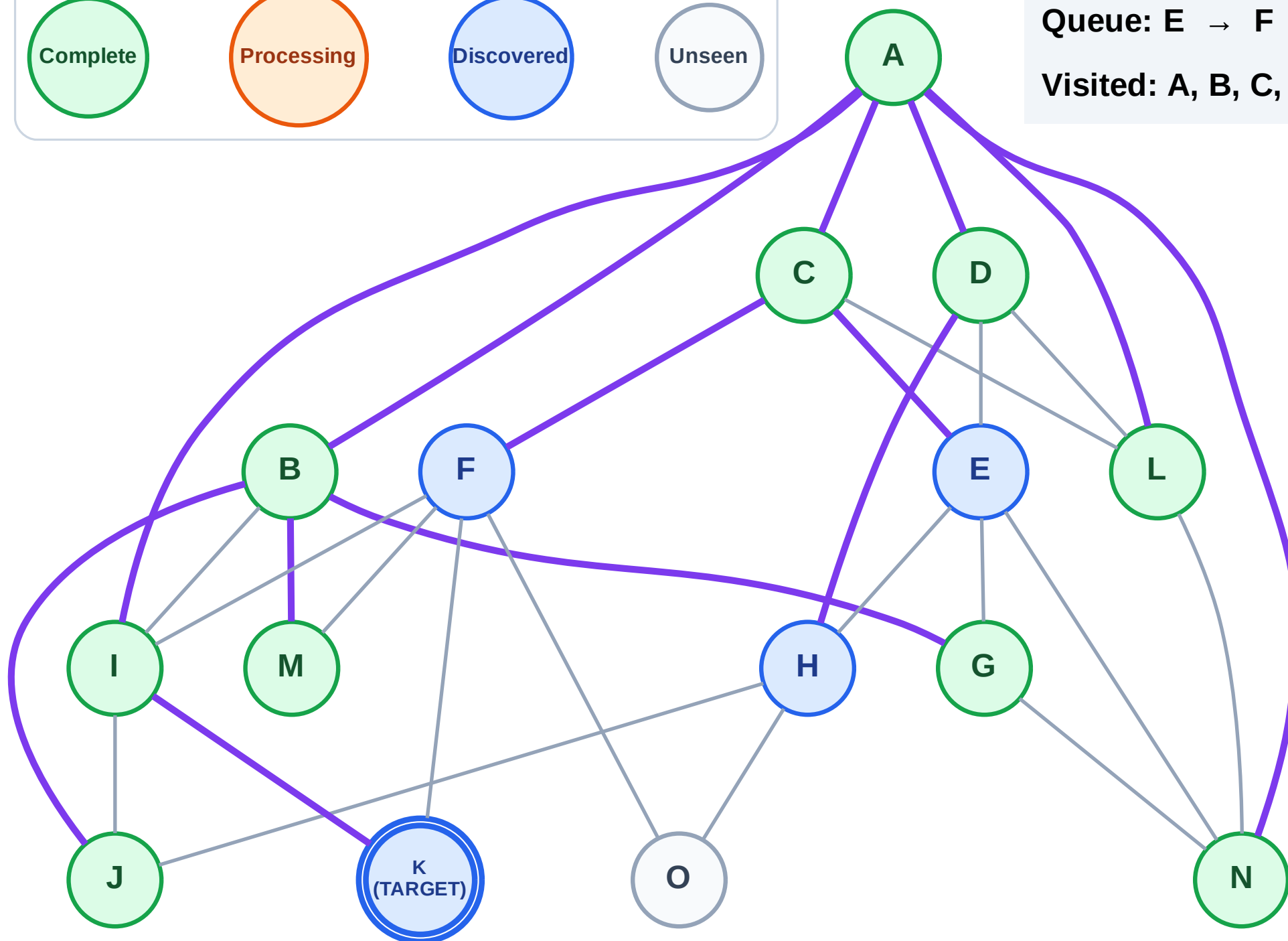
Step 21: No new neighbors from M

Legend



Queue: E → F → H → K

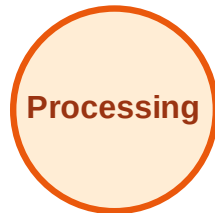
Visited: A, B, C, D, G, I, J, L, M, N



BFS Traversal

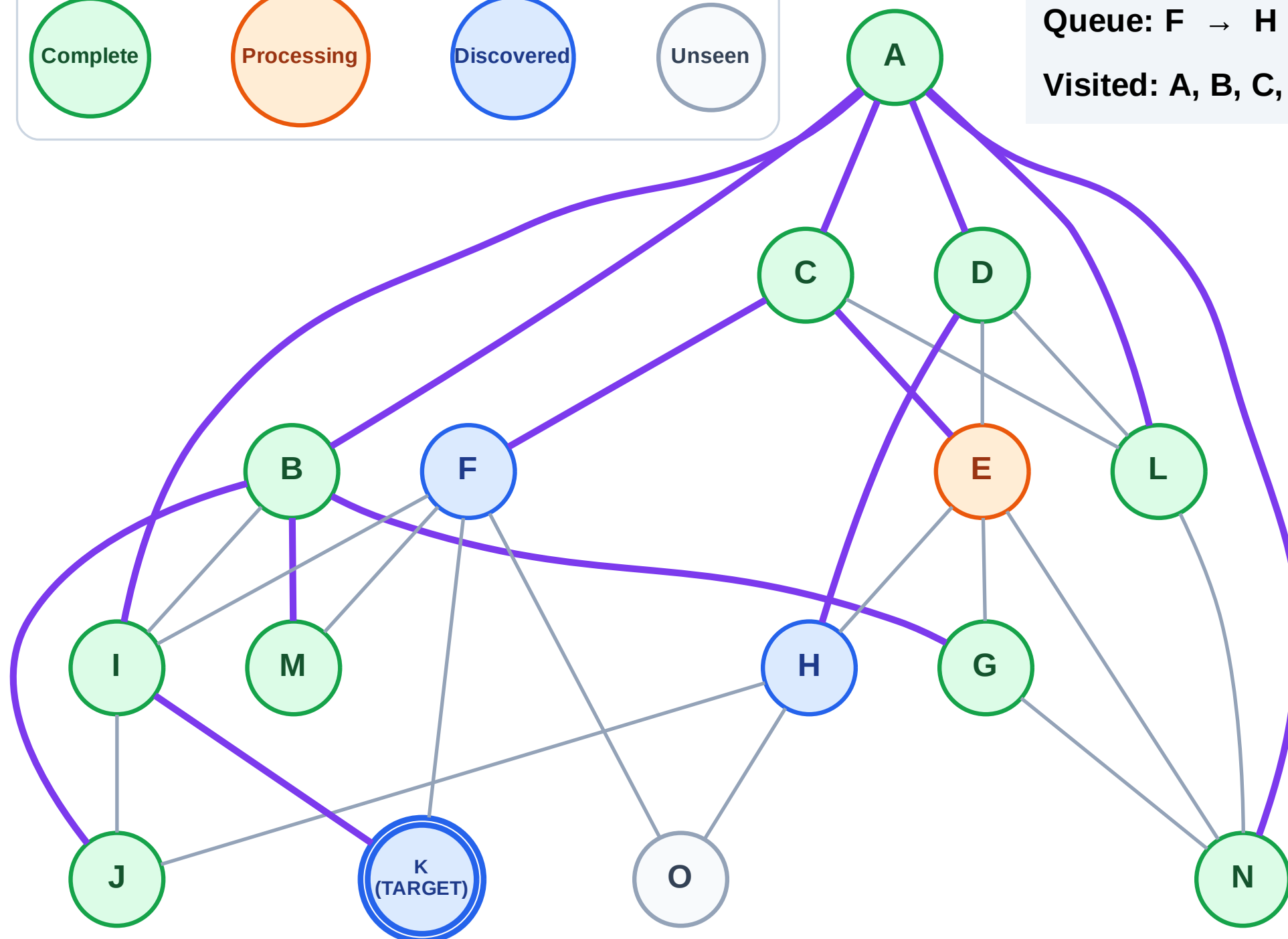
Step 22: Process node E

Legend



Queue: F → H → K

Visited: A, B, C, D, G, I, J, L, M, N



BFS Traversal

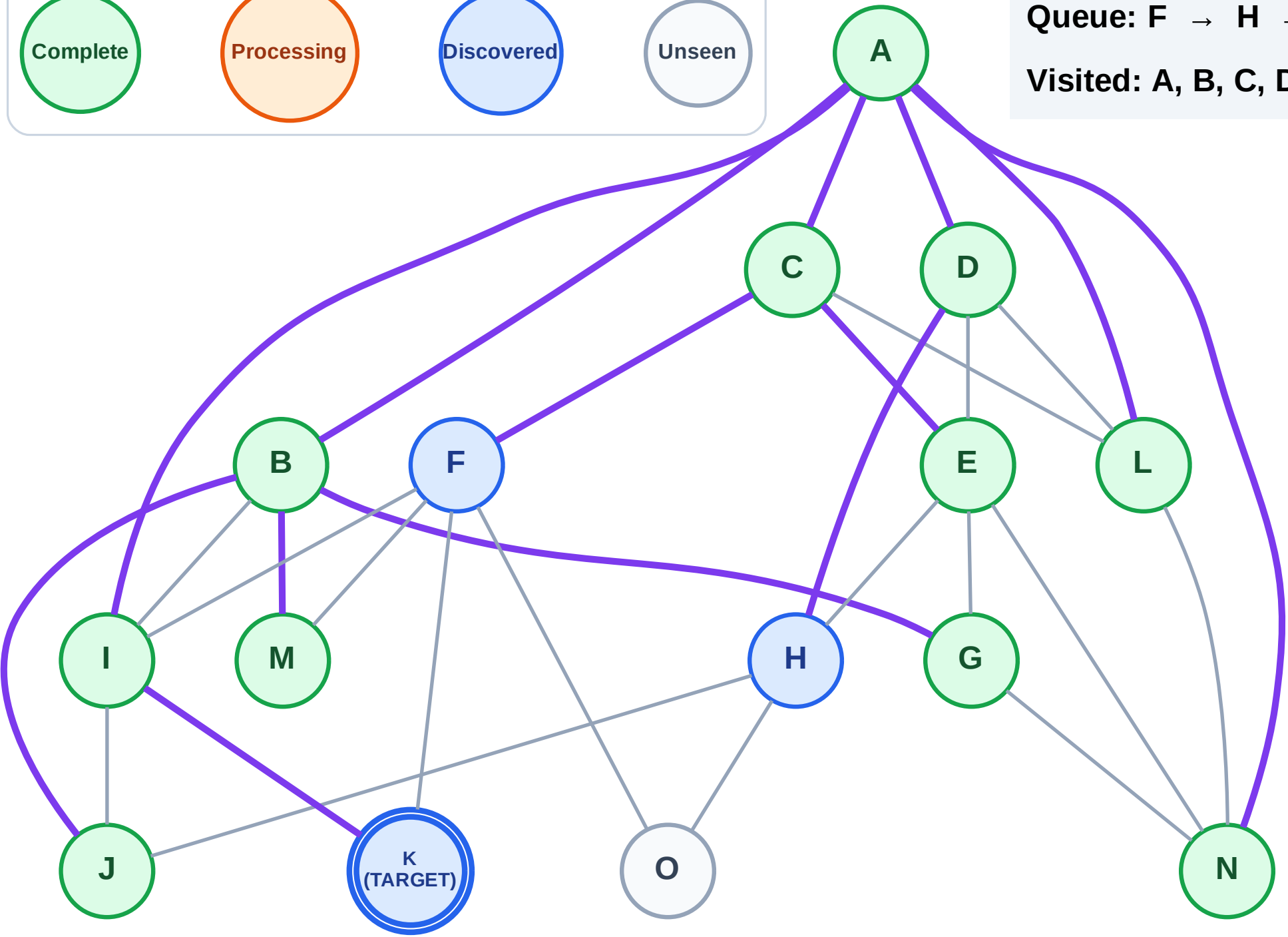
Step 23: No new neighbors from E

Legend



Queue: F → H → K

Visited: A, B, C, D, E, G, I, J, L, M, N



BFS Traversal

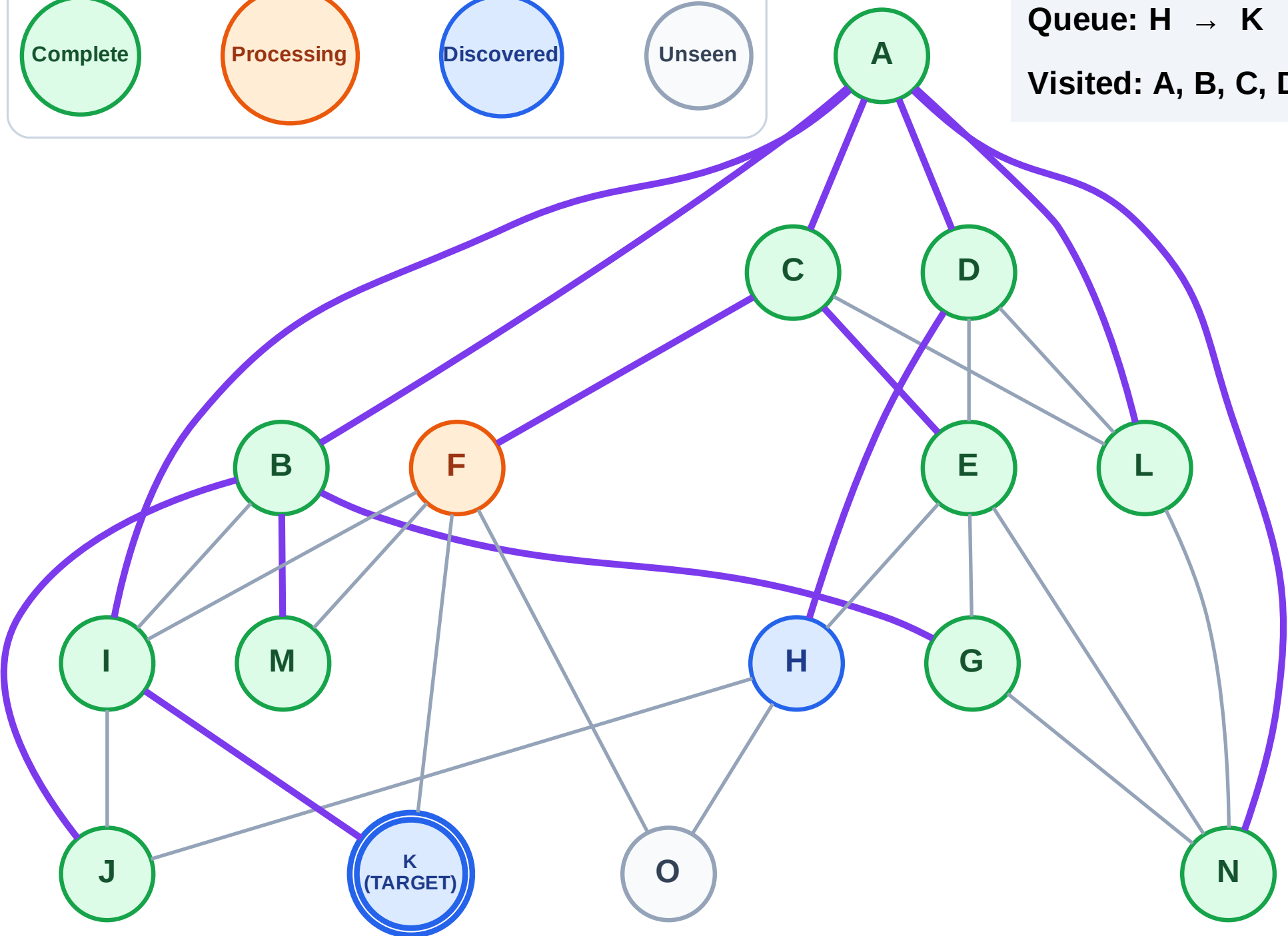
Step 24: Process node F

Legend



Queue: H → K

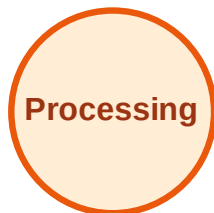
Visited: A, B, C, D, E, G, I, J, L, M, N



BFS Traversal

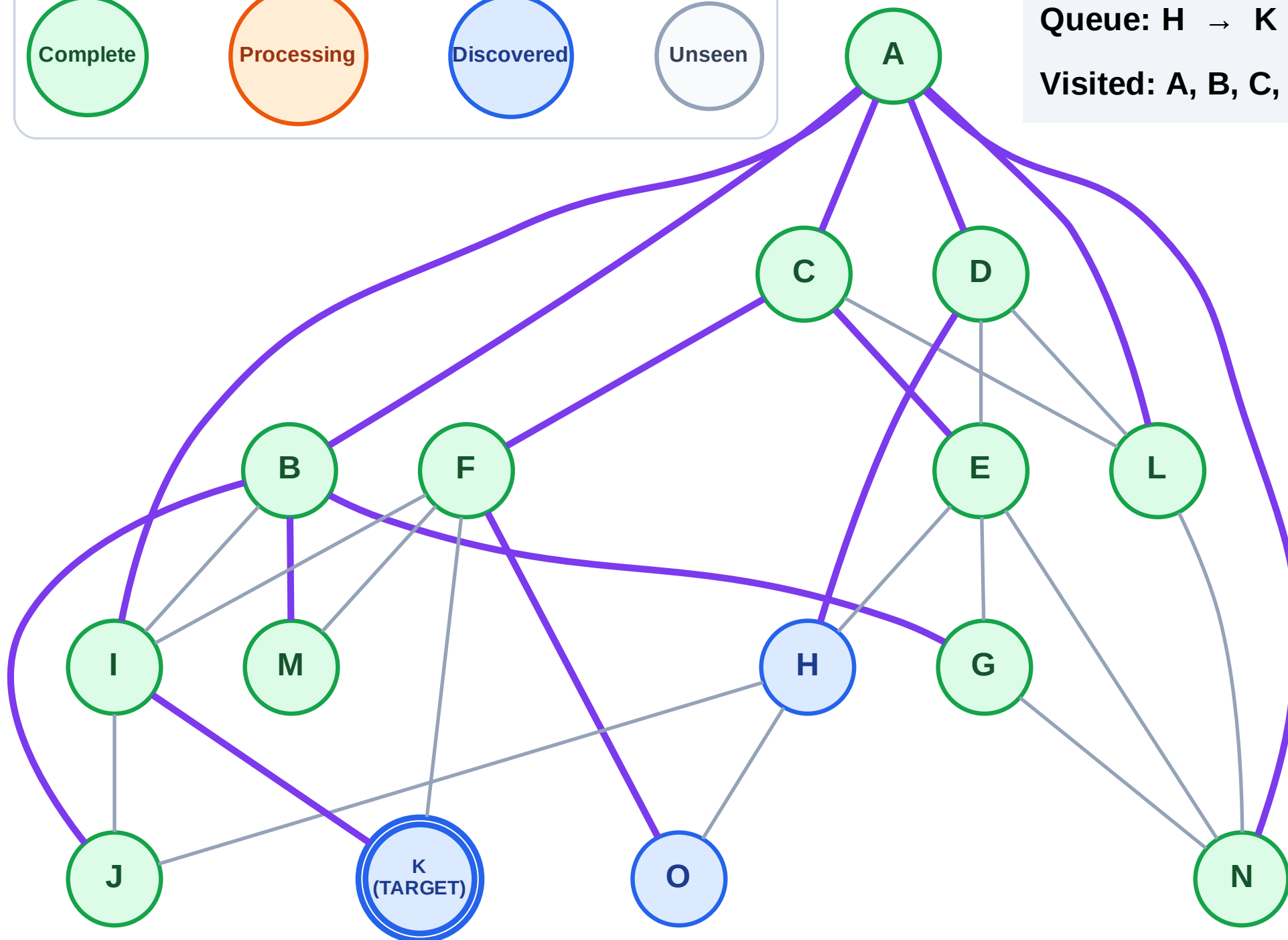
Step 25: Discover O from F

Legend



Queue: H → K → O

Visited: A, B, C, D, E, F, G, I, J, L, M, N



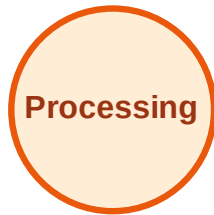
BFS Traversal

Step 26: Process node H

Legend



Complete



Processing



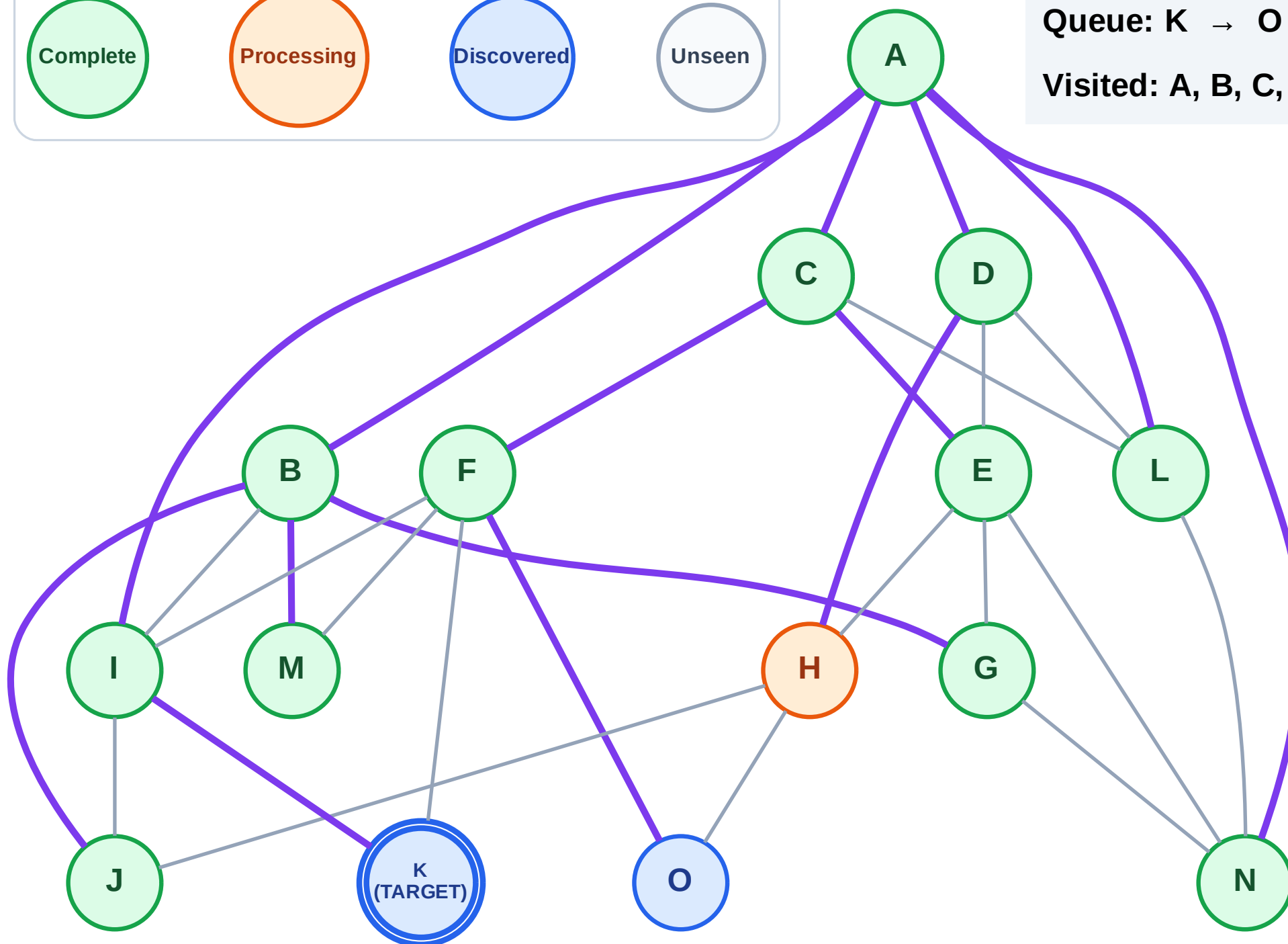
Discovered



Unseen

Queue: K → O

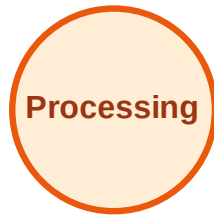
Visited: A, B, C, D, E, F, G, I, J, L, M, N



BFS Traversal

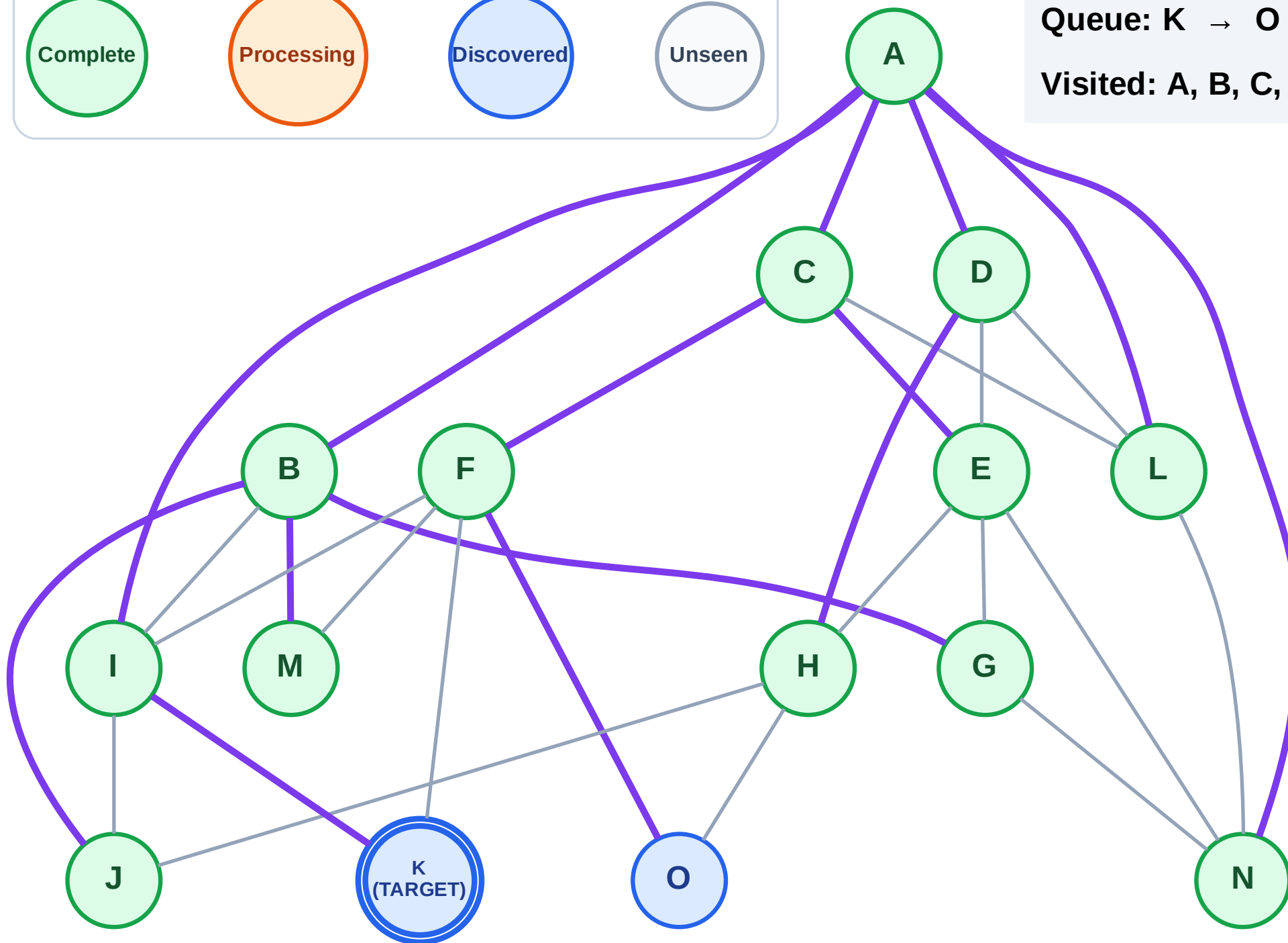
Step 27: No new neighbors from H

Legend



Queue: K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



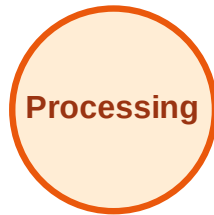
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



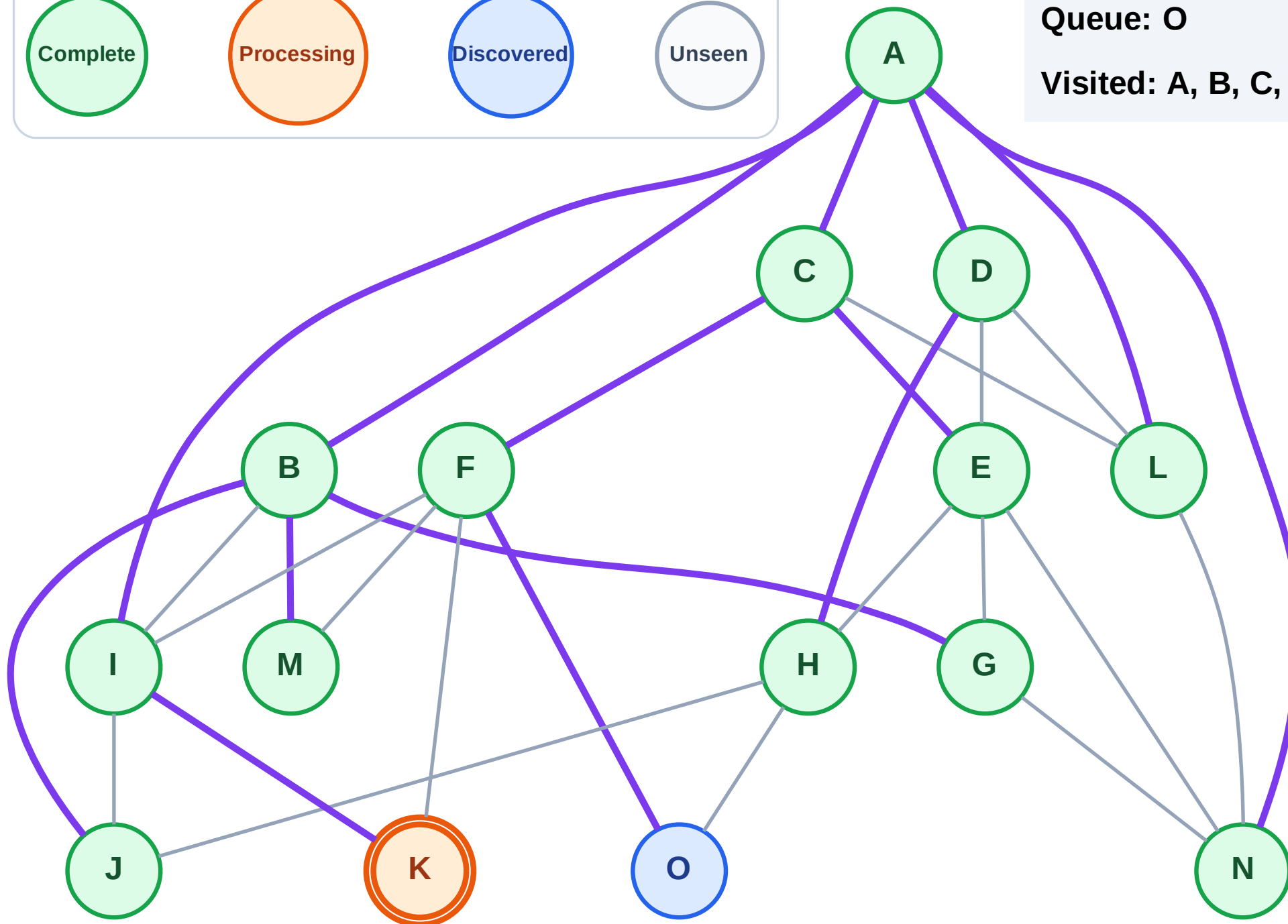
Discovered



Unseen

Queue: O

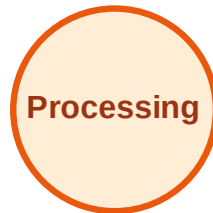
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

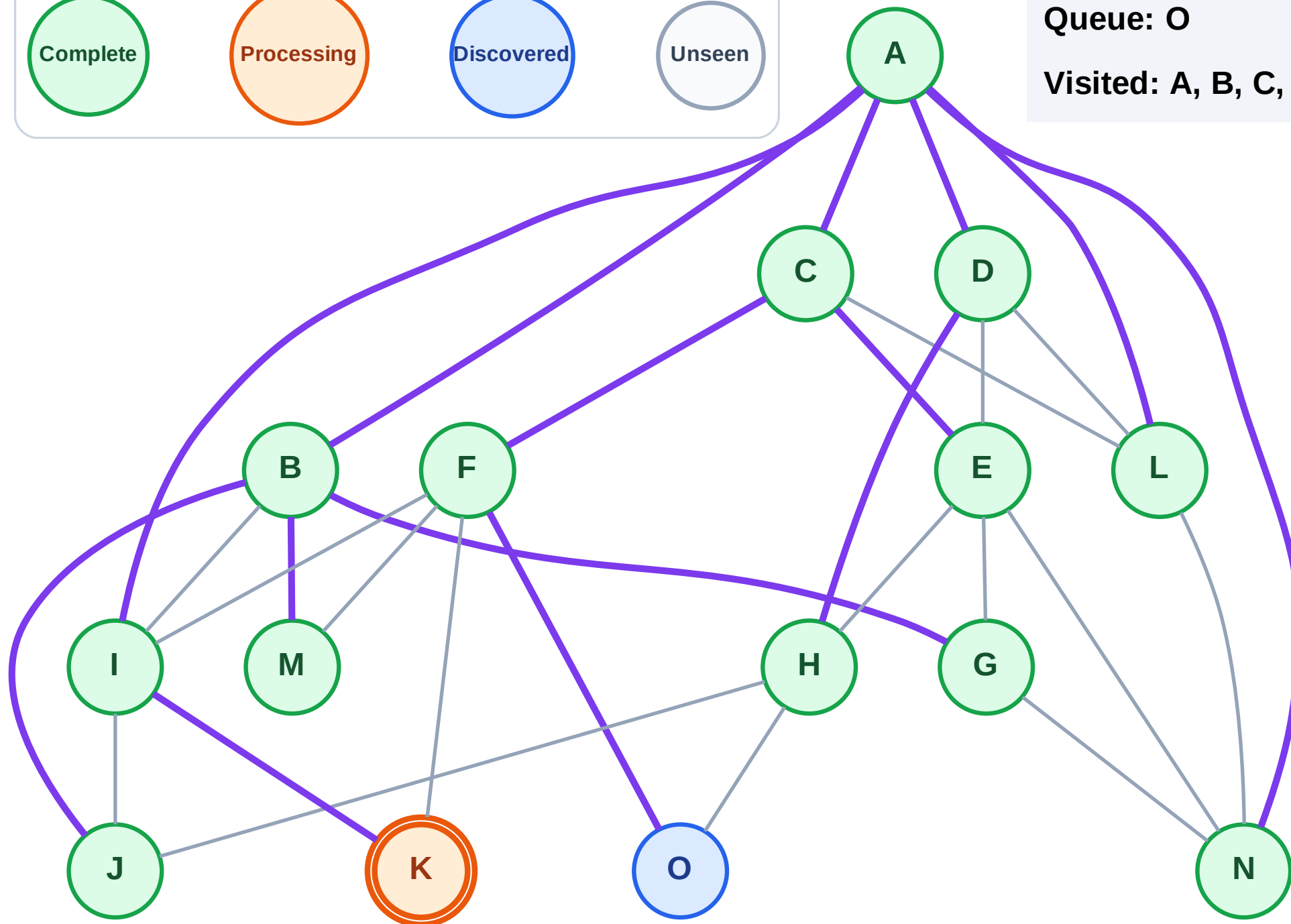
Step 29: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend



Queue: O

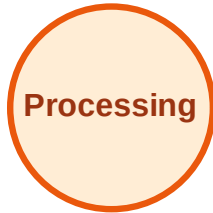
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

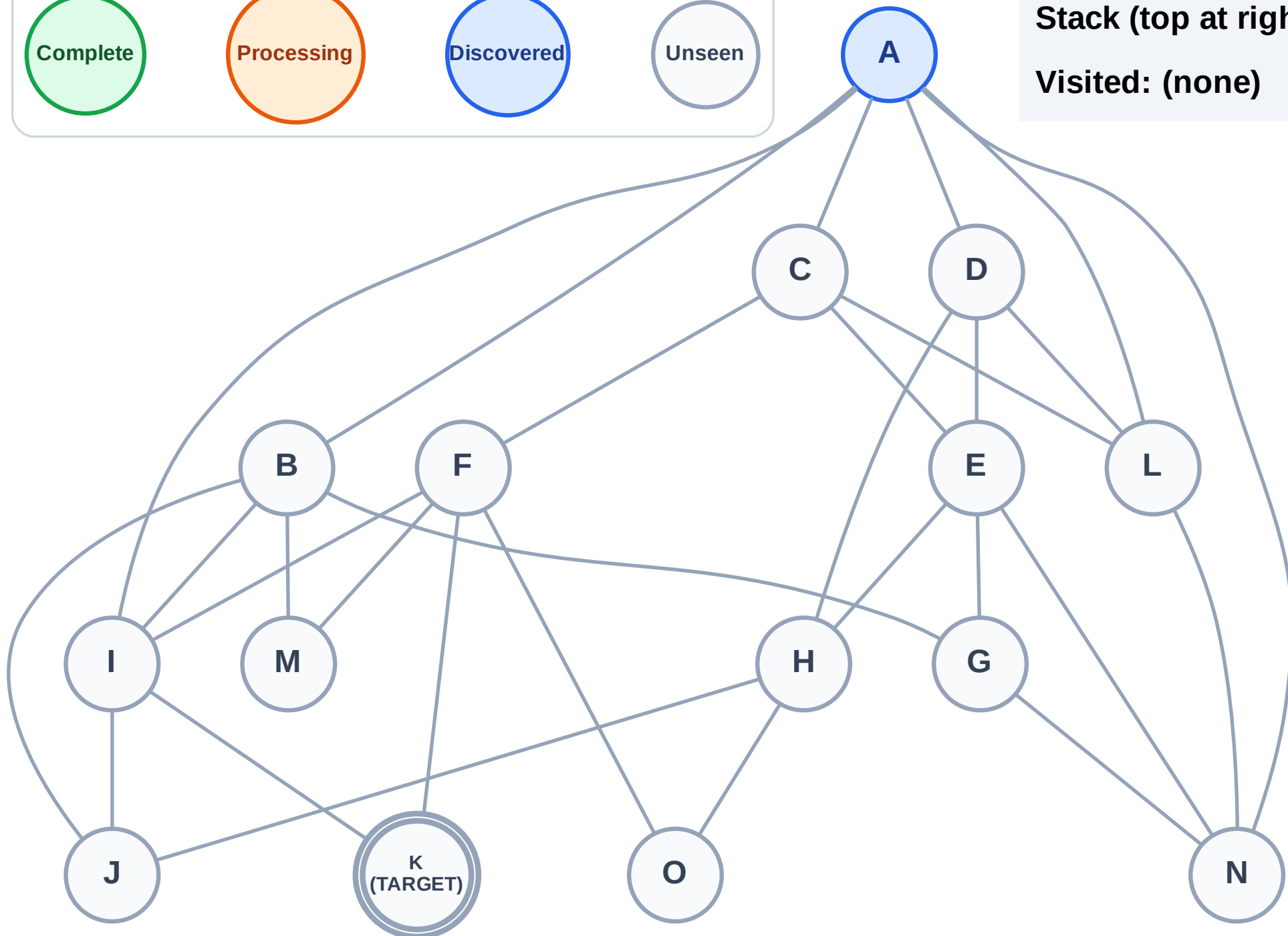
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

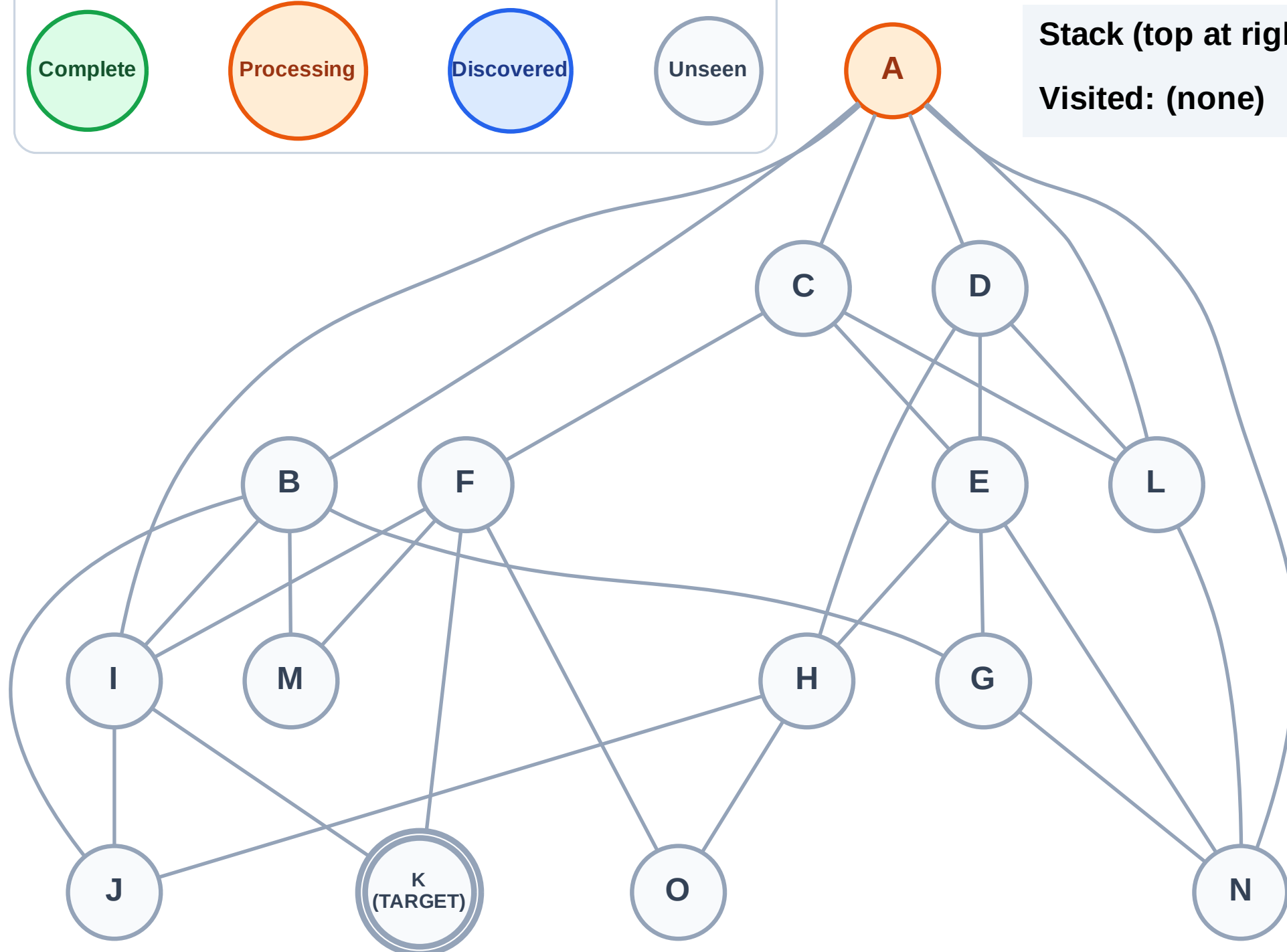
Processing

Discovered

Unseen

Stack (top at right): (empty)

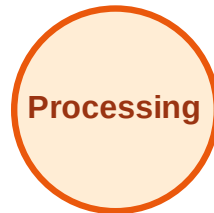
Visited: (none)



DFS Traversal

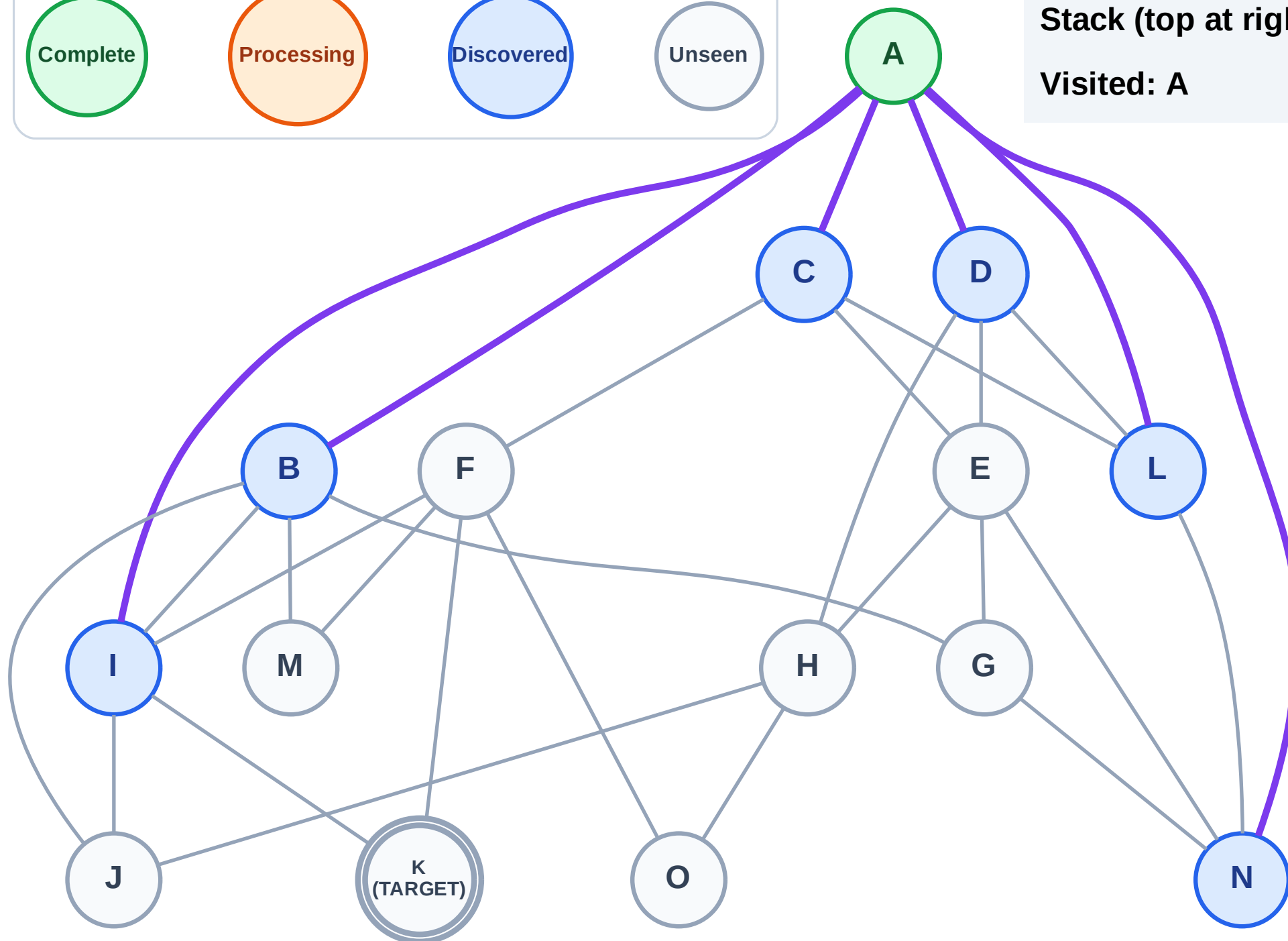
Step 3: Discover N, L, I, D, C, B from A

Legend



Stack (top at right): N | L | I | D | C | B

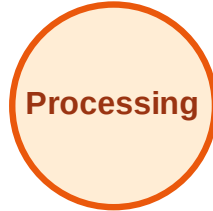
Visited: A



DFS Traversal

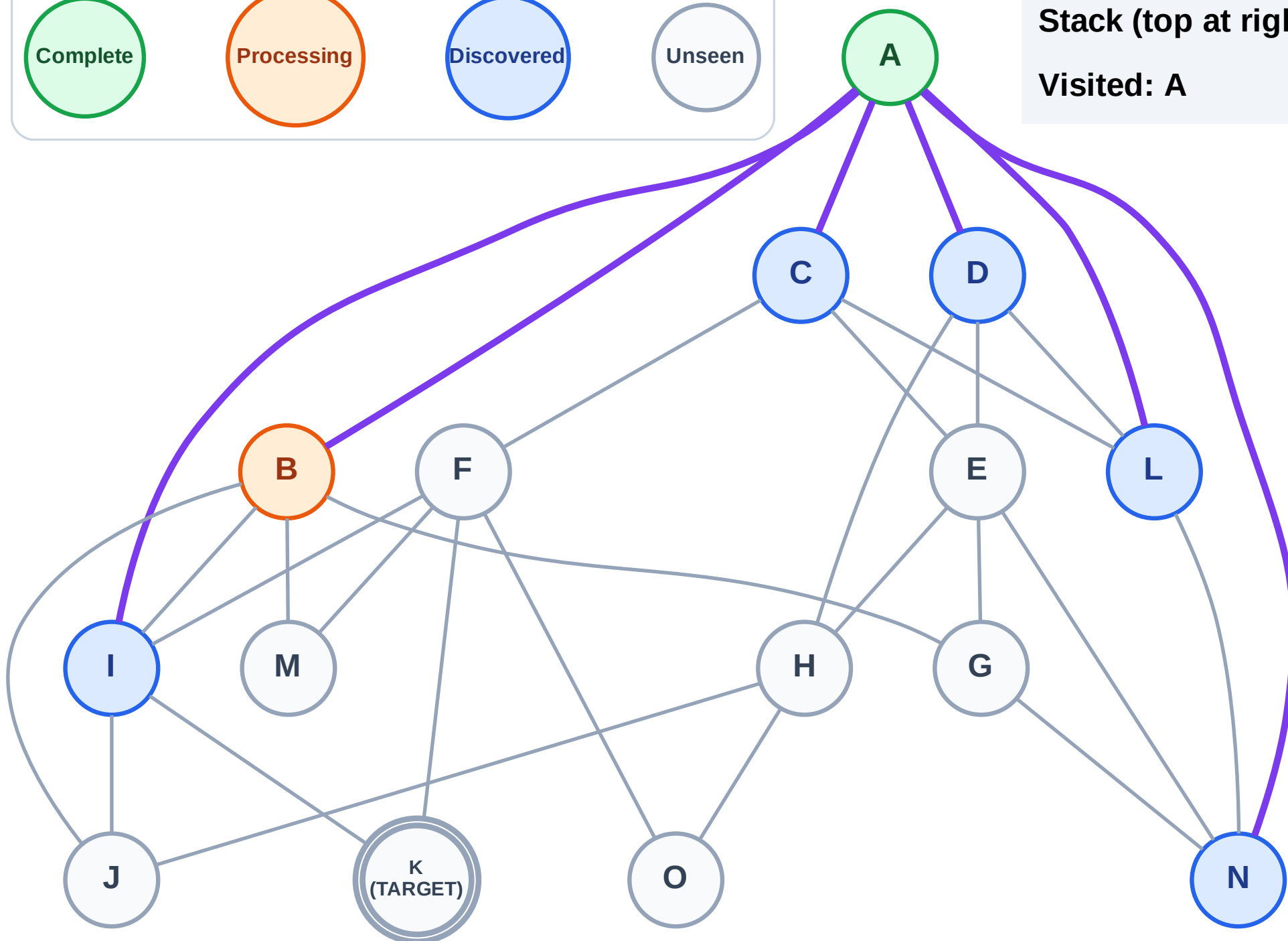
Step 4: Process node B

Legend



Stack (top at right): N | L | I | D | C

Visited: A



DFS Traversal

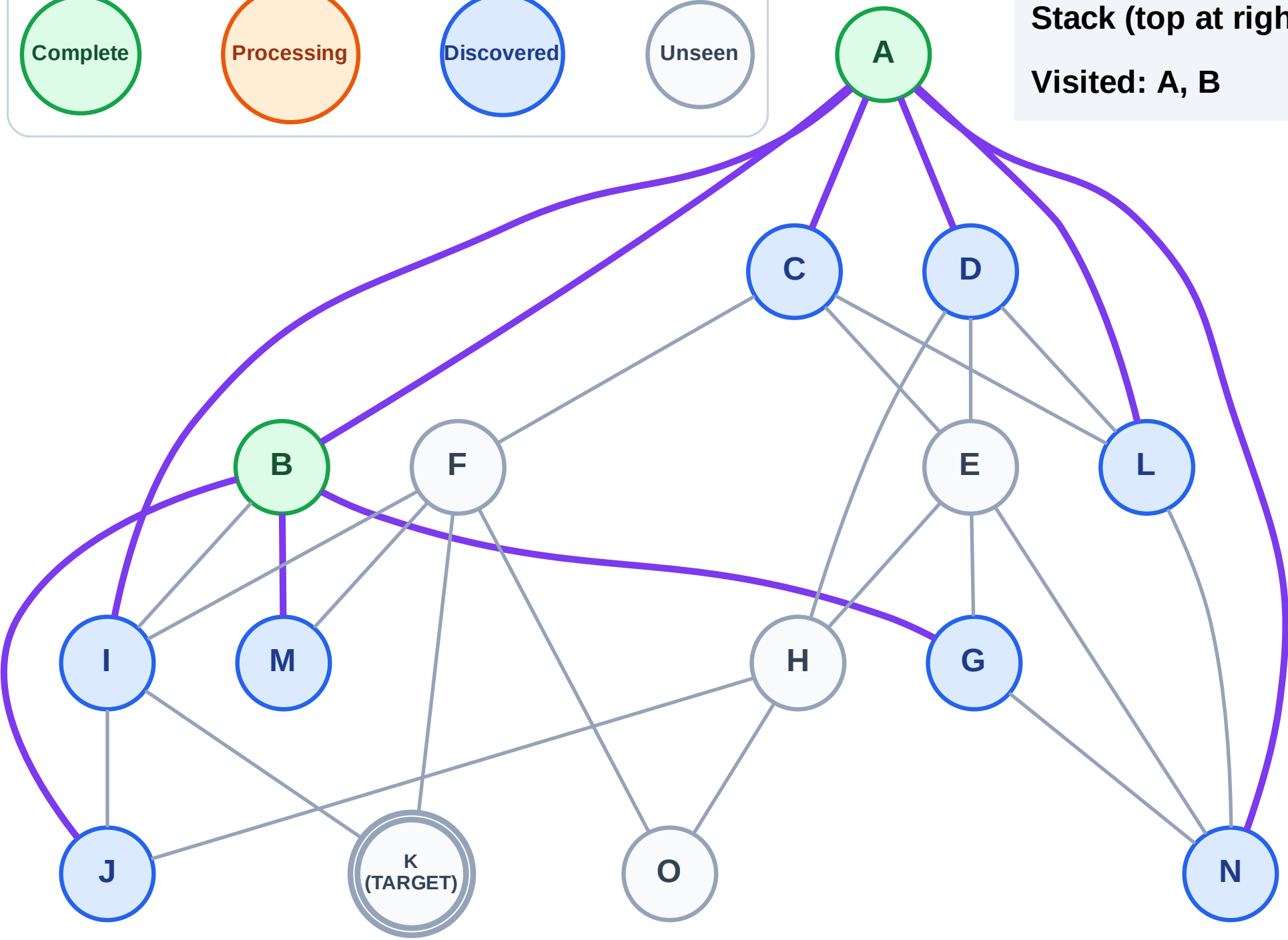
Step 5: Discover M, J, G from B

Legend



Stack (top at right): N | L | I | D | C | M | J | G

Visited: A, B



DFS Traversal

Step 6: Process node G

Legend

Complete

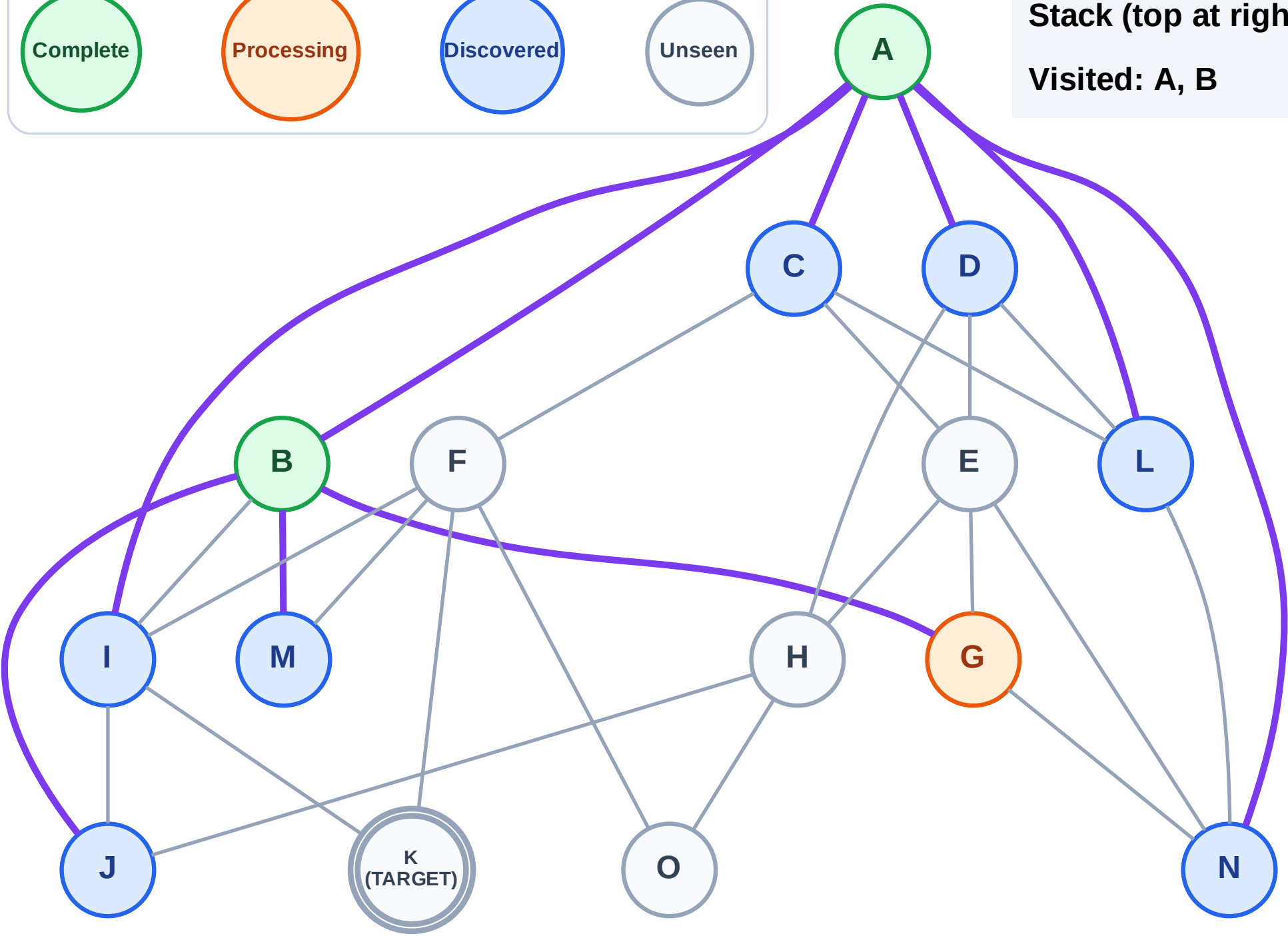
Processing

Discovered

Unseen

Stack (top at right): N | L | I | D | C | M | J

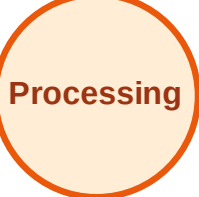
Visited: A, B



DFS Traversal

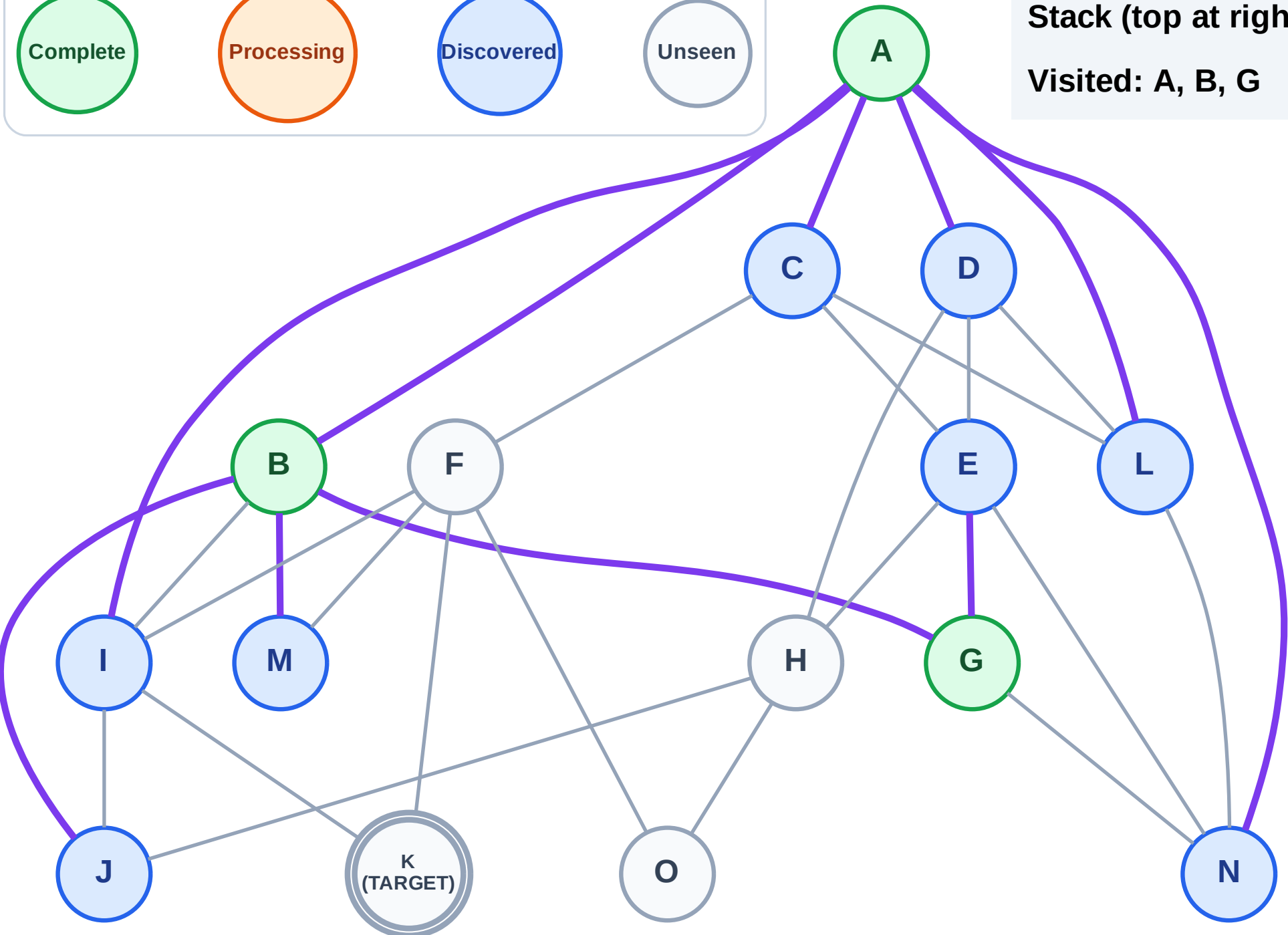
Step 7: Discover E from G

Legend



Stack (top at right): N | L | I | D | C | M | J | E

Visited: A, B, G



DFS Traversal

Step 8: Process node E

Legend

Complete

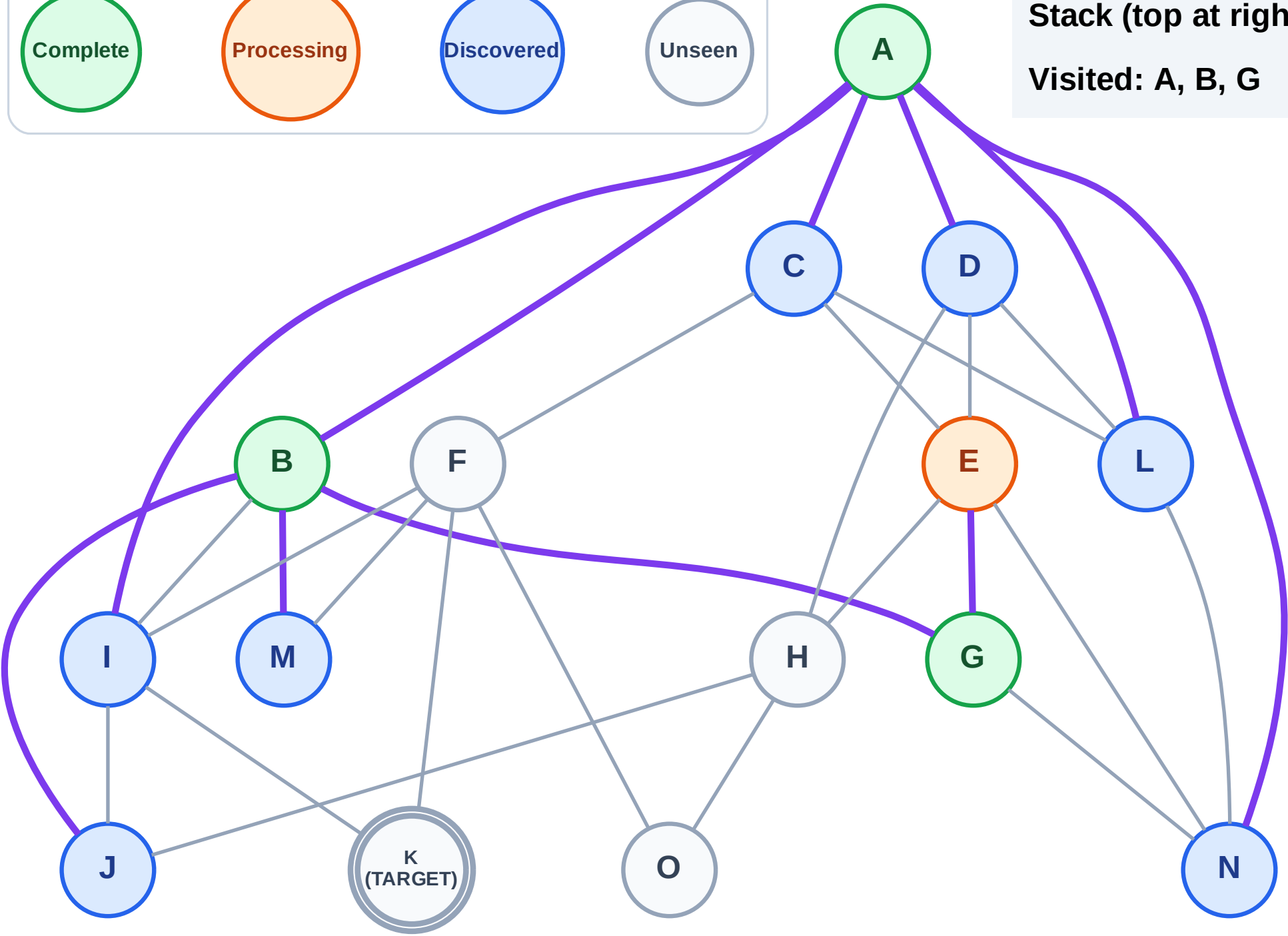
Processing

Discovered

Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, G



DFS Traversal

Step 9: Discover H from E

Legend



Complete



Processing



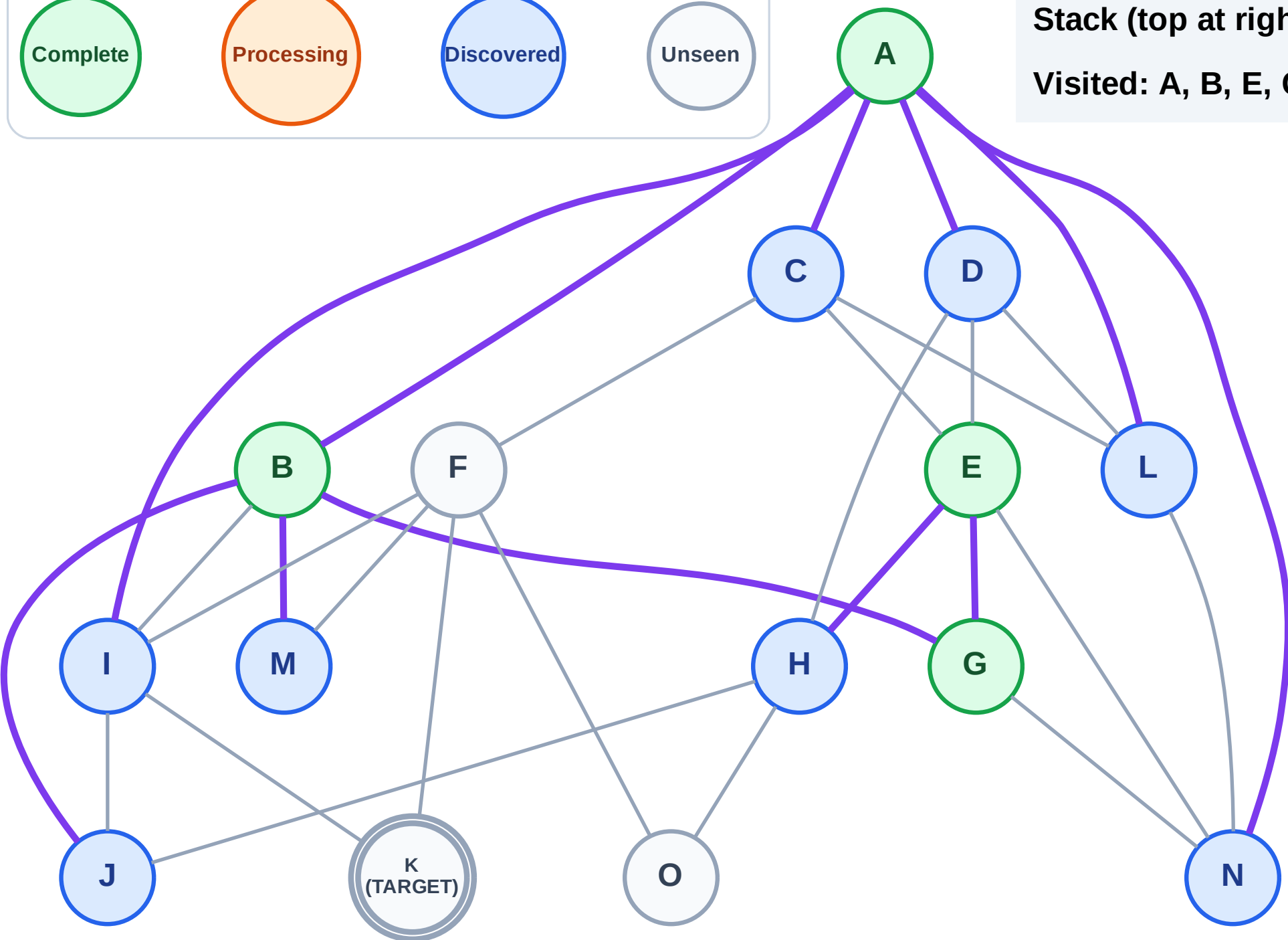
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J | H

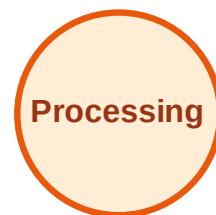
Visited: A, B, E, G



DFS Traversal

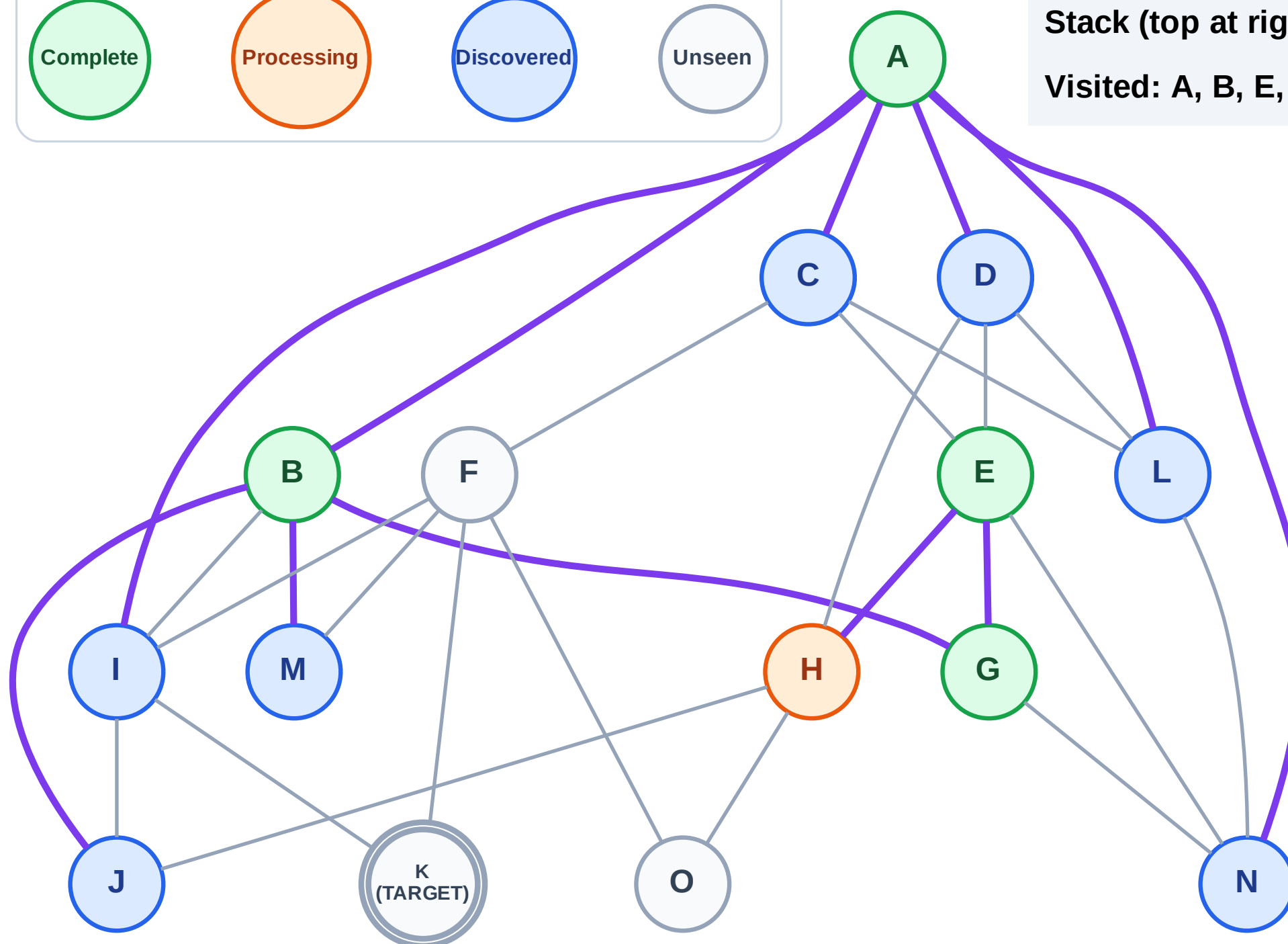
Step 10: Process node H

Legend



Stack (top at right): N | L | I | D | C | M | J

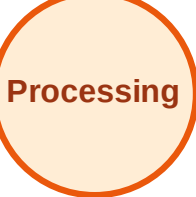
Visited: A, B, E, G



DFS Traversal

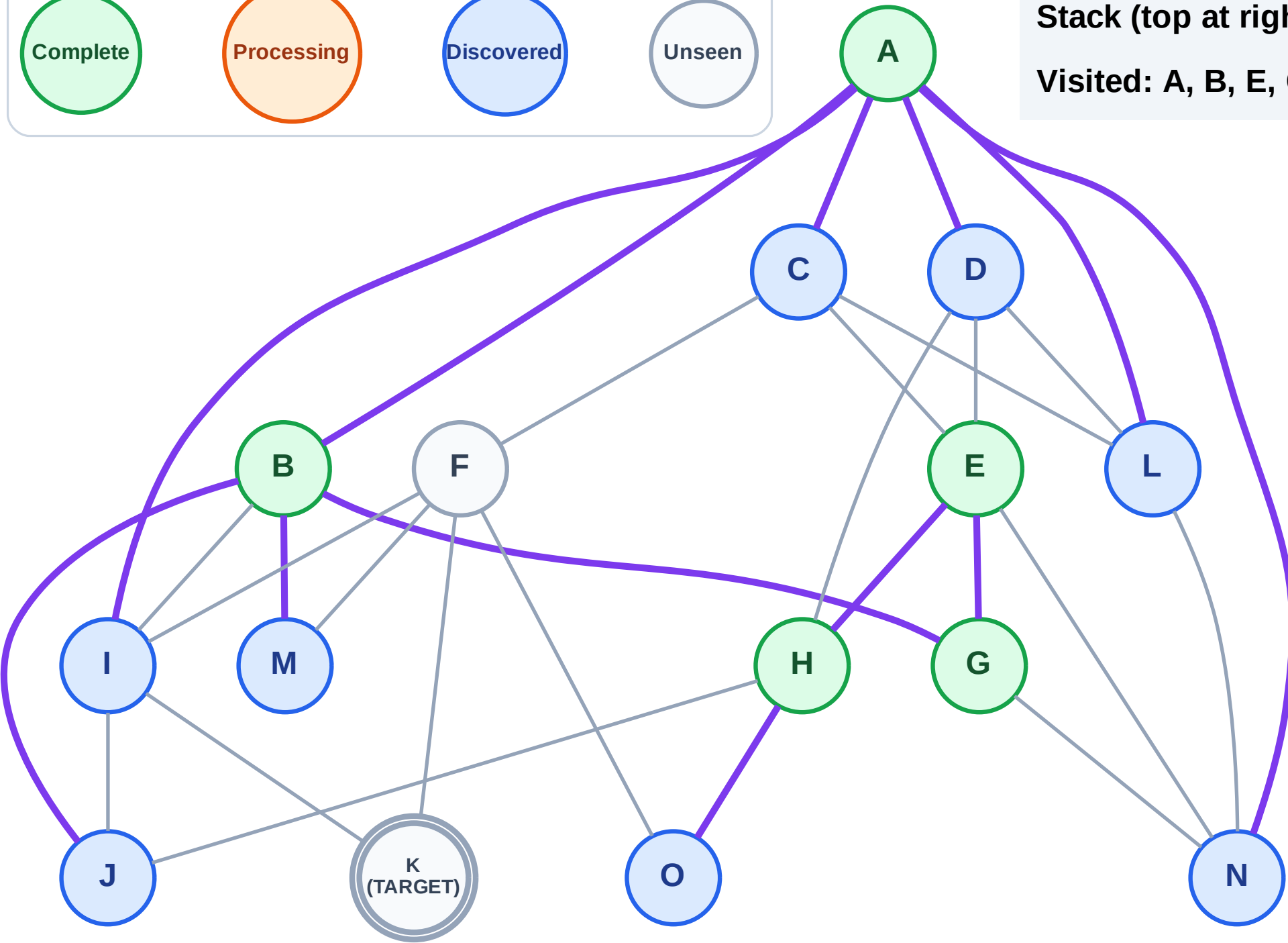
Step 11: Discover O from H

Legend



Stack (top at right): N | L | I | D | C | M | J | O

Visited: A, B, E, G, H



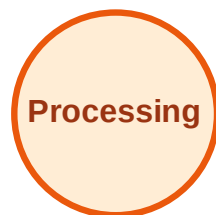
DFS Traversal

Step 12: Process node O

Legend



Complete



Processing



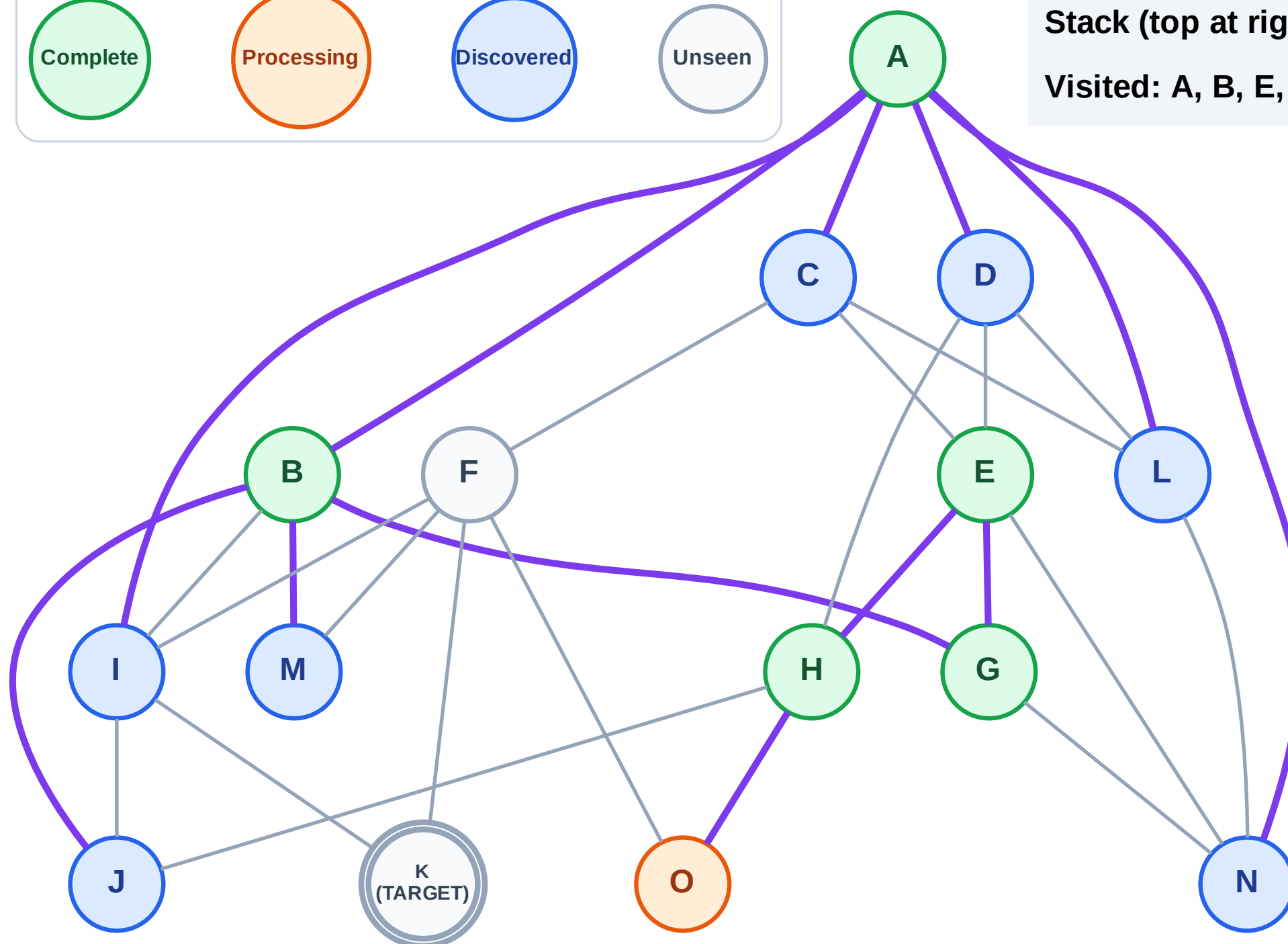
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, G, H



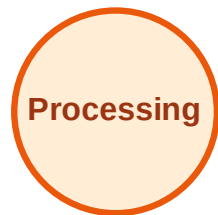
DFS Traversal

Step 13: Discover F from O

Legend



Complete



Processing



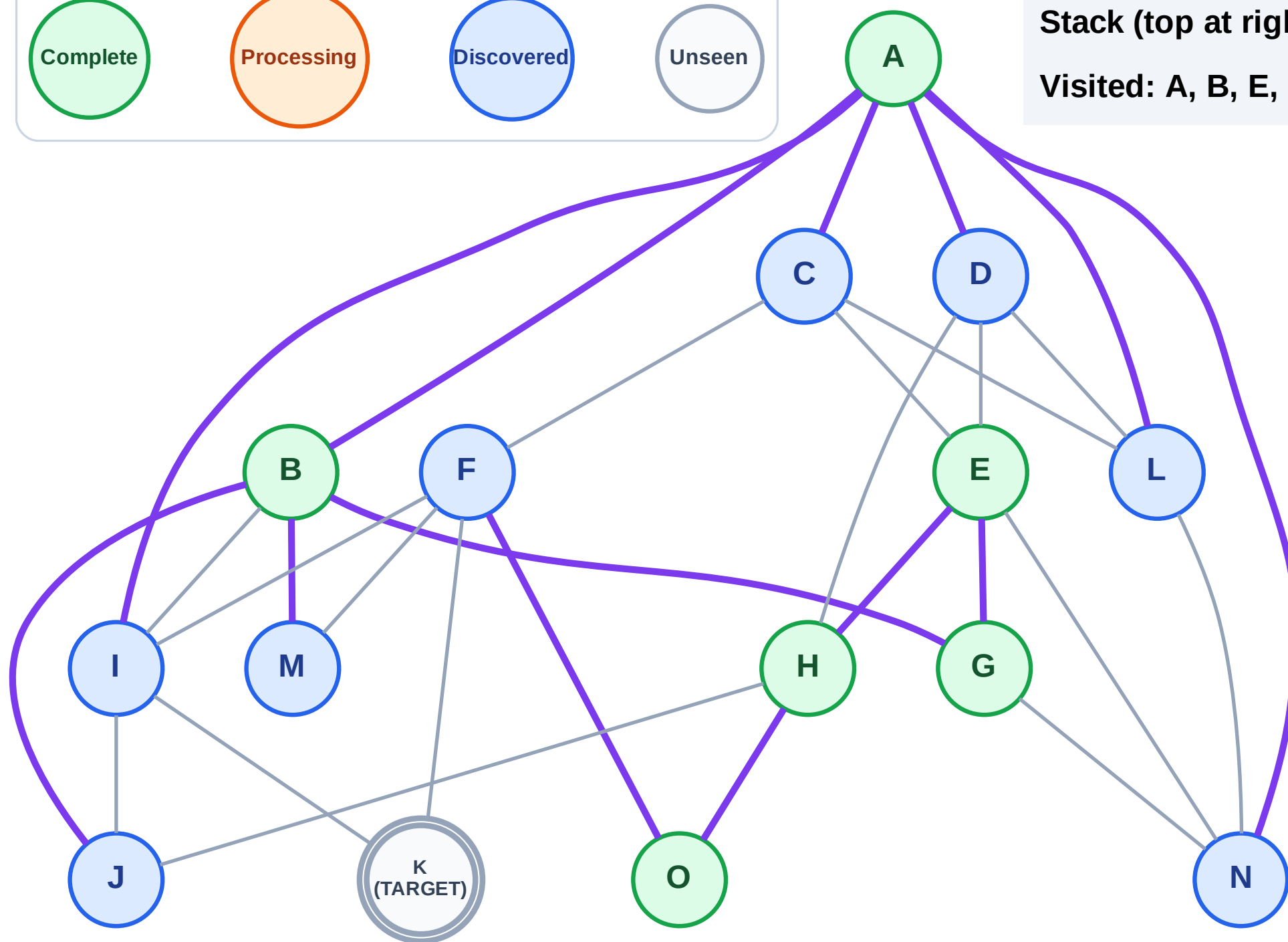
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J | F

Visited: A, B, E, G, H, O



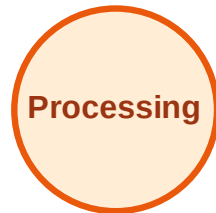
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



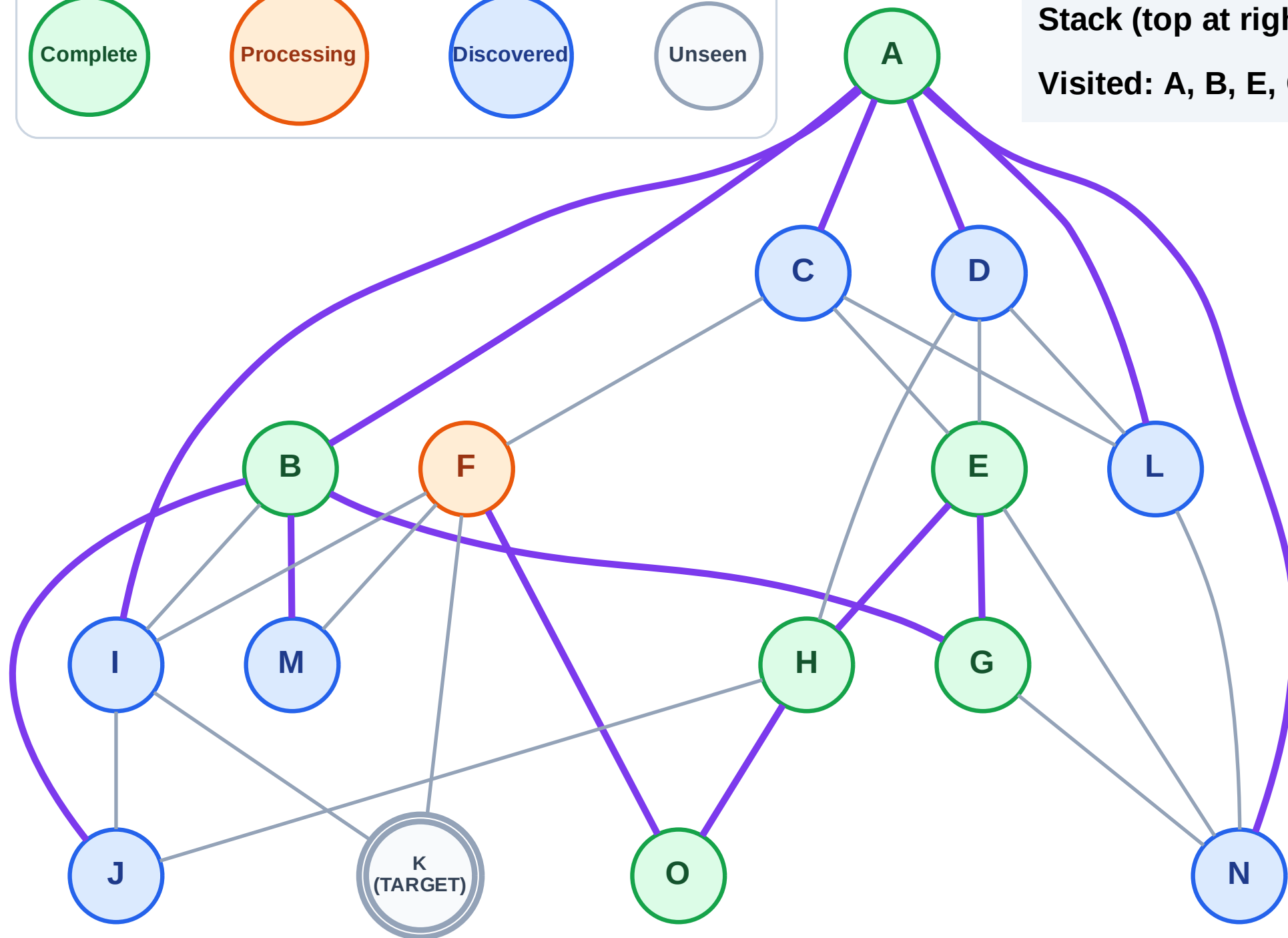
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, G, H, O



DFS Traversal

Step 15: Discover K from F

Legend

Complete

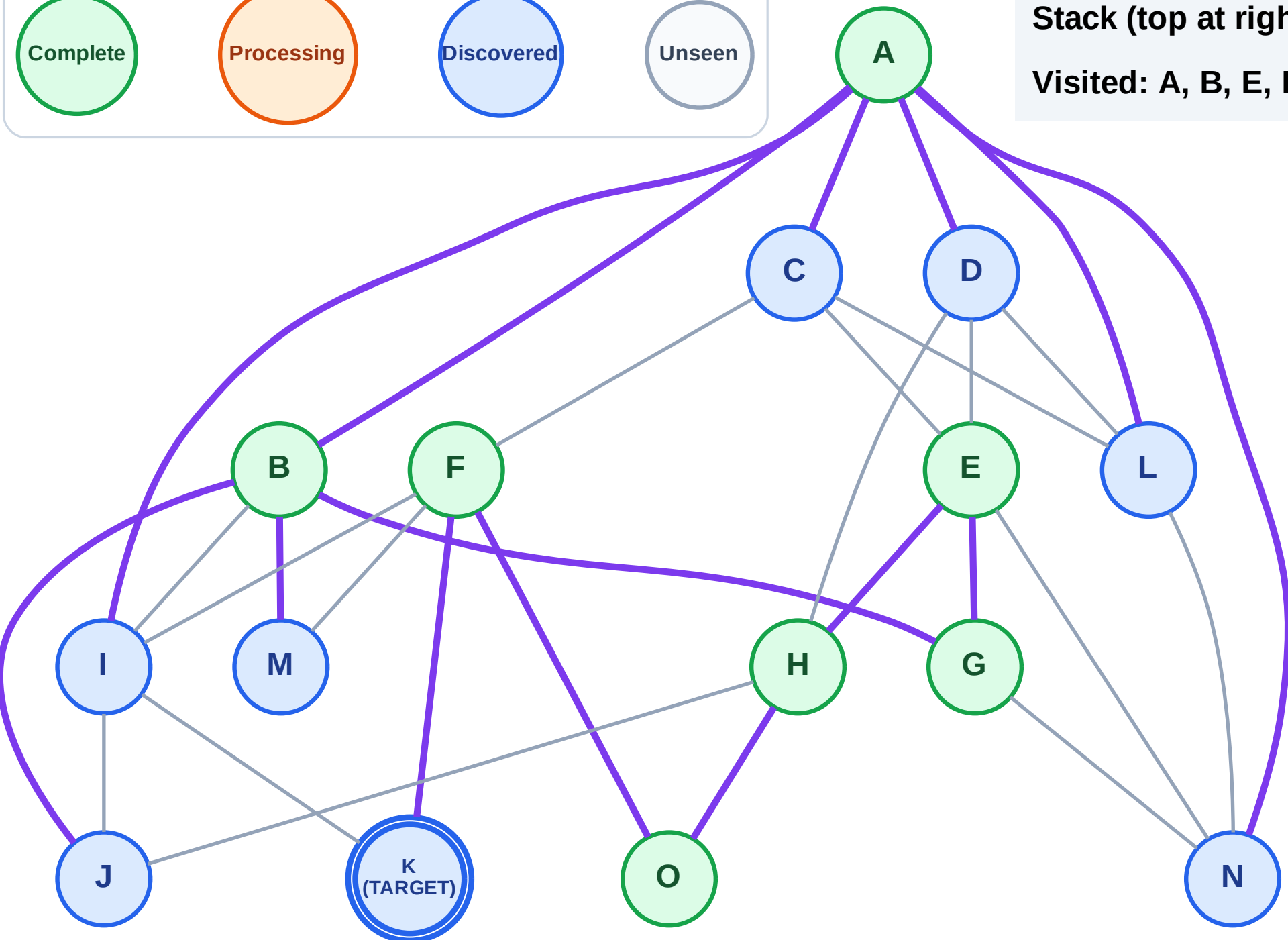
Processing

Discovered

Unseen

Stack (top at right): N | L | I | D | C | M | J | K

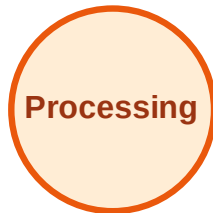
Visited: A, B, E, F, G, H, O



DFS Traversal

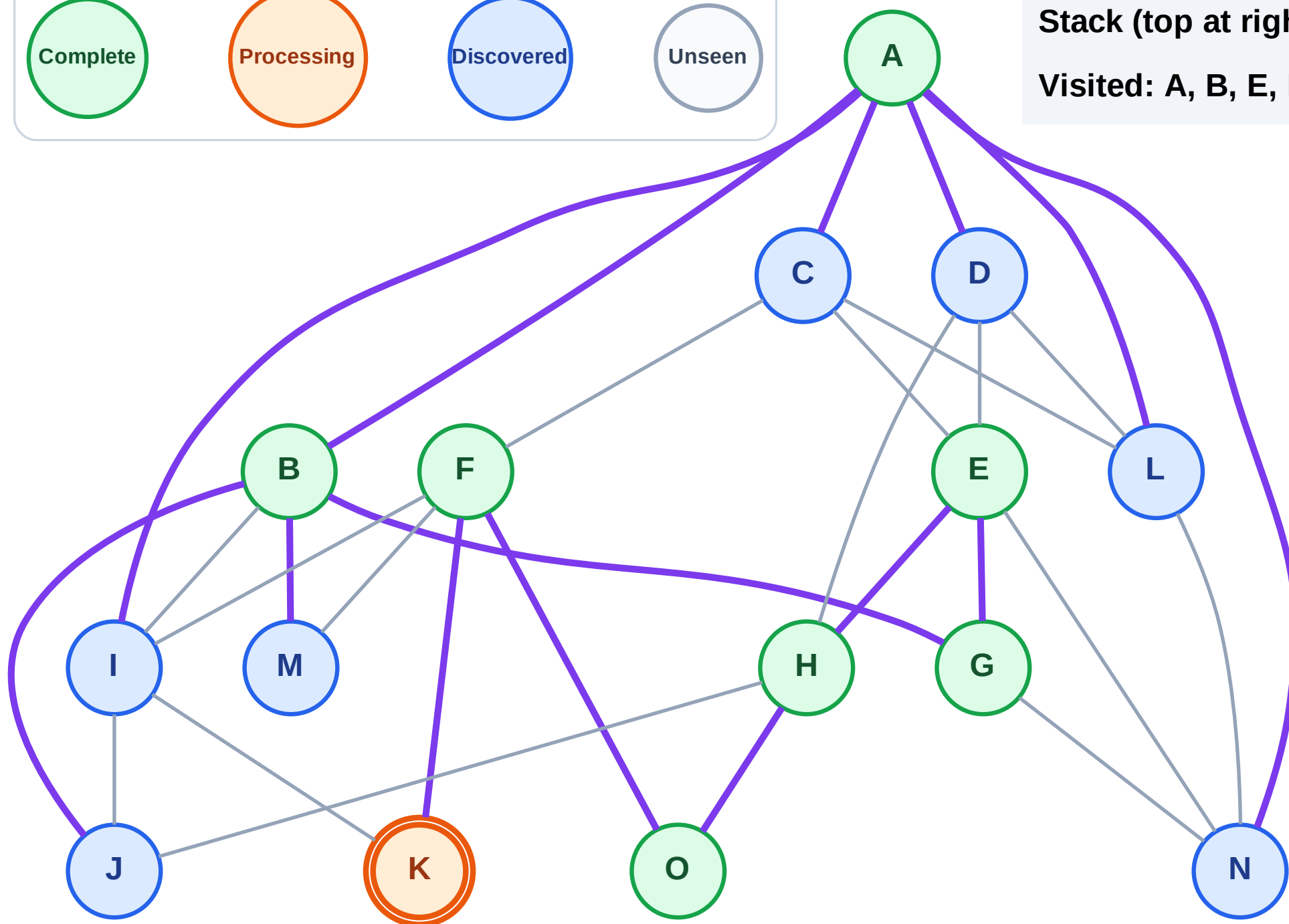
Step 16: Process node K

Legend



Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, F, G, H, O



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (7 hops): A → B → G → E → H → O → F → K

Legend



Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, F, G, H, K, O

